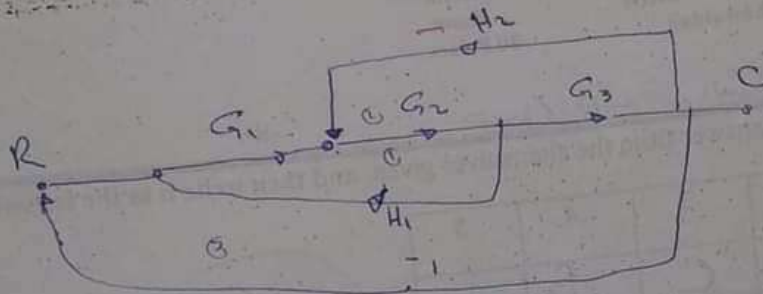


b- Convert the block diagram in (a) to signal flow graph and then, find the transfer function by Mason's gain formula.



$$P = G_1 G_2 G_3$$

$$L_1 = G_1 G_2 (-H_1)$$

$$L_2 = G_2 G_3 (+H_2)$$

$$L_3 = -G_1 G_2 G_3$$

2 loop non touch

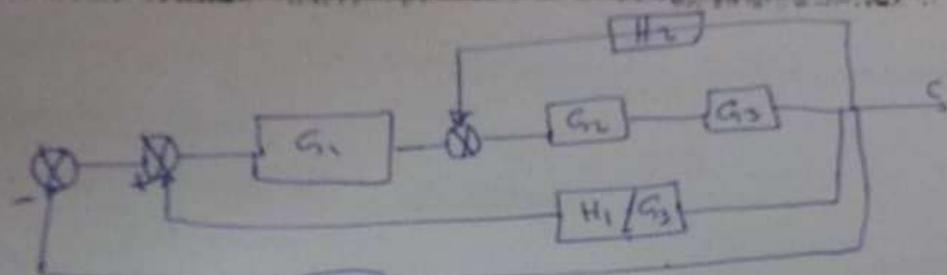
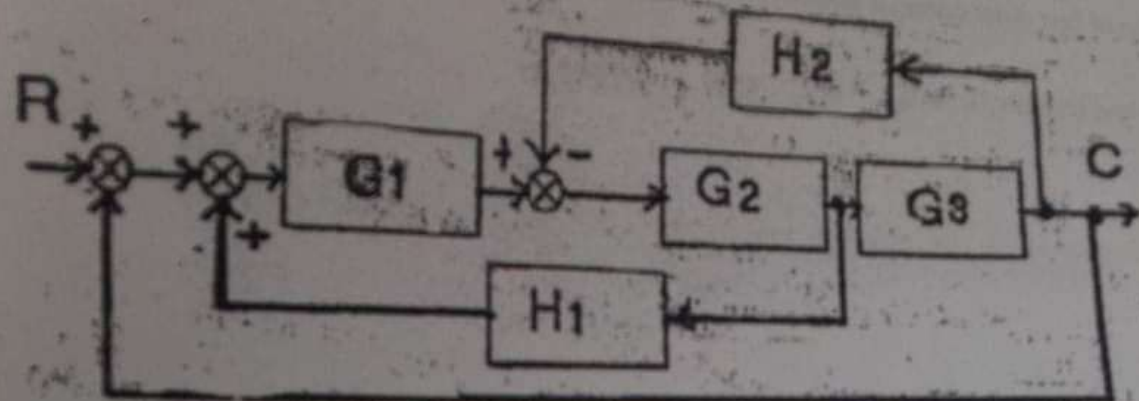
$$\Delta = 1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3$$

$$\Delta = 1$$

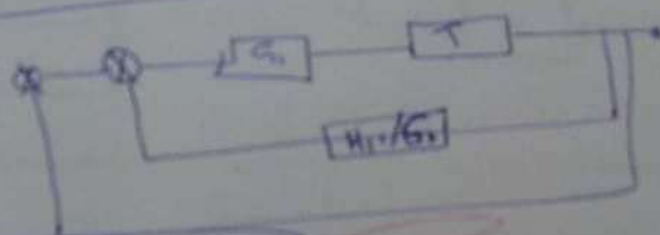
$$T_F = \frac{\Delta_i P_i}{\Delta}$$

$$= \frac{1 \cdot G_1 G_2 G_3}{1 + G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3}$$

a- Reduce the following block diagram to find the transfer function of the system.



$$T = \frac{G_2 \cdot G_3}{1 + G_2 G_3 H_1}$$



$$T_2 = \frac{T \cdot G_1}{1 + T \cdot G_1 \cdot \frac{H_2}{G_3}}$$

$$= \frac{G_1 \cdot G_2 \cdot G_3}{1 + G_2 G_3 G_1 H_1 + H_2 - G_2}$$

$$T_E = \frac{T_2}{1 + T_2}$$

$$G_2 G_3 G_1$$

$$1 + G_2 G_3 G_1 H_1 H_2 - G_1 G_1 H_1 + G_2 G_3 G_1$$

Select the most correct answer from the alternatives given, and then write it in the table below.

1	2	3	4	5	6	7	8	9	10
C	C	B	B A	D	B	C	B	B	A

- 1- The major advantages of closed-loop control systems is
 A- Simple construction
 B- Ease of maintenance.
 C- High accuracy
 D- There is no stability problem

2- The partial fraction expansion of $Y(s) = \frac{10(s+1)}{s(s^2+4s+13)}$ is $\frac{a}{s} + \frac{bs+c}{s^2+4s+13}$

A) $Y(s) = \frac{a}{s} + \frac{b_1}{s^2+4s+13}$

B) $Y(s) = \frac{a}{s} + \frac{b_1 s}{s^2+4s+13}$

C) $Y(s) = \frac{a}{s} + \frac{b_1 s + b_2}{s^2+4s+13}$

D) $Y(s) = \frac{a}{s} + \frac{b_1}{s^2+4} + \frac{b_2}{s+13}$

3- The solution of the differential equation $y''(t) + 4y'(t) + 3y(t) = \delta(t)$, where $\delta(t)$ is a unit impulse, is

A- $y(t) = 0.5e^{-t} + 0.5e^{-3t}$

B) $y(t) = 0.5e^{-t} - 0.5e^{-3t}$

C- $y(t) = 0.5e^{-3t} - 0.5e^{-t}$

D- $y(t) = e^{-t} + e^{-3t}$. (Note that $y'(0)=0$ and $y(0)=0$)

4- The state space representation of the following system is

$$5\frac{d^3y}{dt^3} + 10\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 2y = u(t).$$

A)

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -0.4 & -1 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0.2 \end{bmatrix} u$$

$$y = [1 \ 0 \ 0] \mathbf{x}$$

B) $\mathbf{x}' = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ -0.4 & -1 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} u$

$$y = [1 \ 0 \ 0] \mathbf{x}$$

C) $\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ +0.4 & +1 & +2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$

$$y = [1 \ 0 \ 0] \mathbf{x}$$

~~D) $\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -0.4 & -1 & -2 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u$~~

$$y = [1 \ 0 \ 0] \mathbf{x}$$

5- single-input, single-output system has the state variable representation

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ -5 & -10 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = [0 \ 10] \mathbf{x}$$

The transfer function of the system $T(s) = Y(s)/U(s)$ is:

Circle the most accurate answer for the following statements (20 marks):

1- One of the following statements is not true for open loop control system

a- The output has no effect on the control action.

☒ b- Stability is a major problem.

☐ c- It is easier to build than a closed loop control system

d- None.

2- One of the following statements is not true for control system transfer function:

a- It is independent of the magnitude and nature of the input

b- The transfer functions of many physically different systems can be identical

☒ c- It provides information concerning the physical structure of the system

d- None

3- The number of state variables in spring-mass-dashpot translation mechanical system is

A.4

B.3

C.2

☒ D.1

4- The inverse Laplace transform of $Y(s) = \frac{s+3}{s(s^2+6s+10)}$ is

A) $0.3 - 0.3e^{-3t} \cos t - e^{-3t} \sin t$ ☒ B) $0.3 - 0.3e^{-3t} \cos t - 0.1e^{-3t} \sin t$

C) $0.1 - 0.3e^{-3t} \cos t - 0.1e^{-3t} \sin t$ ☒ D) $0.3 - 0.3e^{-3t} \cos t - \sin t$

5- Transfer function approach usually used when the system is

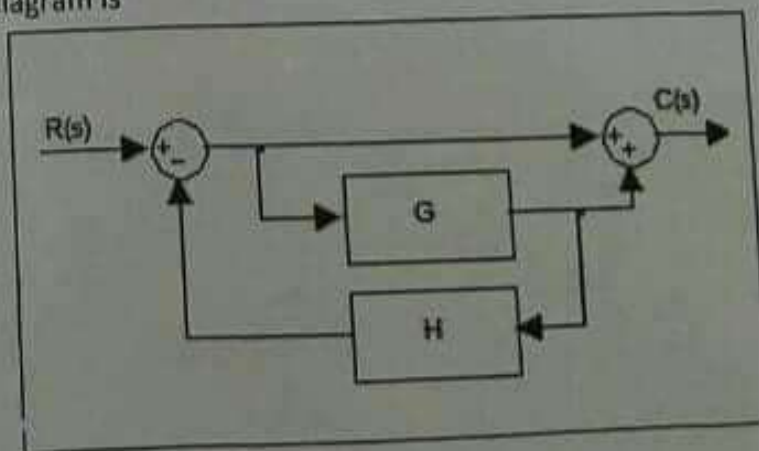
A) single input multi output B) multi input single output

☒ C) single input single output D) multi input multi output

6- The transfer function for the shown block diagram is

A) $\frac{1+G}{1-GH}$ B) $\frac{1+G}{1+GH}$

C) $\frac{G}{1+GH}$ D) none



Select the most correct answer from the alternatives given, and then write it in the table below.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	B	C	D	B	B	C	D	D	B	X	B	X	0	C

1. For closed control system which of the following statements is correct?

- (A) Less expensive
(B) May be unstable stable ✓
(C) Construction is simple and maintenance easy
(D) Errors are caused by disturbances

2. The first control system developed is

- (A) Temperature control system (B) level control system
(C) speed control system (D) robot system ✓

3. Transfer function of a system is used to calculate which of the following?

- (A) The order of the system (B) The time constant
(C) The output for any given input (D) The steady state gain

4. In temperature control system, the feedback path is

- (A) Thermometer (B) Analog to digital converter (C) Interface (D) All above ✓

5. What should be the value of Laplace transform for the time-domain signal function?

$f(t) = e^{-at} \sin(\omega t) \cdot u(t)$

- (A) $1/(s+a)$ (B) $\omega / \{(s+a)^2 + \omega^2\}$ (C) $(s+a) / \{(s+a)^2 + \omega^2\}$ (D) $A\omega / (s^2 + \omega^2)$

6. When the function $F(s)$ has multiple poles, the type of the function $f(t)$ is

- (A) Exponential (B) damped ramp (C) undamped (D) critically damped sinusoidal

7. The inverse Laplace transform of $F(s) = 2/(s+1)(s+2)$ is

- (A) $2e^{-t} - 2e^{-2t}$ (B) $2e^{-t} + 2e^{-2t}$ (C) $2e^{-t} - 2e^{-2t}$ (D) $2e^{-t} + 2e^{-2t}$

8. The system given by $y''(t) + ay'(t) + y(t) = x(t)$, where a constant, can be represented by

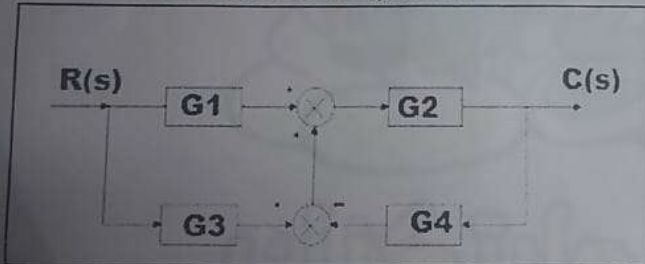
- (A) simulation model (B) transfer function (C) state space model (D) all mentioned

9. The transfer function of RL circuit, where the applied voltage is the input and the current is the output is

- (A) $\frac{Ls}{RLs+1}$ (B) $\frac{s}{RLs+1}$ (C) $\frac{L}{Ls+1}$ (D) $\frac{1}{Ls+R}$

10. Consider the block diagram shown,

If the transfer function of the system is



given by $T(s) = \frac{G1G2+G2G3}{1+X}$. Then X is:

- (A) $G2G3G4$ (B) $G2G4$ (C) $G1G2G4$ (D) $G3G4$

A control system is given as

$$\dot{x}_1'(t) = x_1(t) + x_2(t) + u_1(t) + u_2(t)$$

$$\dot{x}_2'(t) = x_1(t) + u_1(t)$$

$$y_1(t) = x_1(t) + u_2(t)$$

$$y_2(t) = u_1(t) + u_2(t)$$

Answer 12 and 13

12. the state matrix A is

$$A \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$B \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$C \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$D \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

13. the output matrix D is

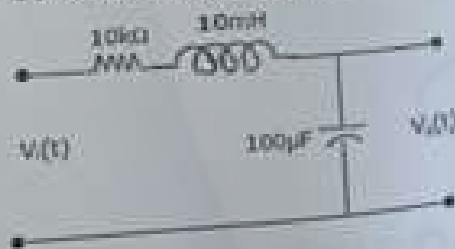
$$A \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$B \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$C \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

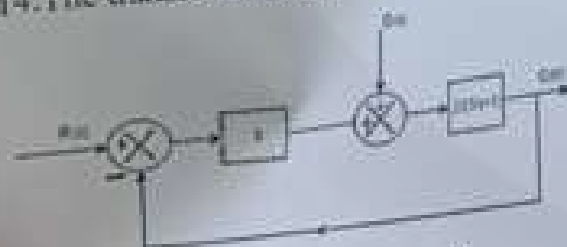
$$D \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

14. For the circuit shown in the Figure below, the initial conditions are zero. Its transfer function $H(s) = V_C/V_i$



- A) $\frac{1}{s^2 + 10^3 s + 10^4}$
 B) $\frac{10^4}{s^2 + 10^3 s + 10^4}$
 C) $\frac{10^4}{s^2 + 10^3 s + 10^4}$
 D) $\frac{10^4}{s^2 + 10^3 s + 10^4}$
 E) None of these

14. The transfer function from D(s) to Y(s) is:



$$\frac{2}{s+7}$$

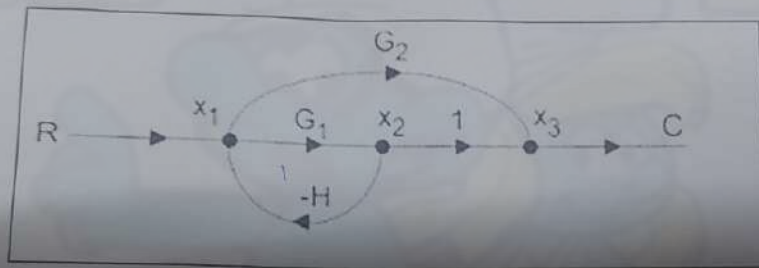
$$b) \frac{2}{3s+1}$$

$$c) \frac{6}{3s+7}$$

$$d) \frac{2}{3s+6}$$

A node which has both incoming and outgoing branches is called a/an
 (A) Input node (B) mixed node (C) output node (D) general node

13. Use Mason formula to calculate the transfer function of the following figure



- (A) $G1/(1+G2H)$ (B) $G2/(1+G1H)$
 (C) $(G1+G2)/(1+G1H)$ (D) None

14. In the given state space equations

$$\dot{x}_1(t) = x_1(t) + x_2(t) + u_1(t) + u_2(t) \quad \dot{x}_2(t) = x_1(t) + u_1(t) \quad \dot{x}_3(t) = x_2(t)$$

the state matrix A is

3x3

- (A) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (B) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (C) $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ (D) None

15. for a standard translation mechanical system, The input signal to the system is
 (A) displacement (B) velocity (C) applied force (D) acceleration

• CO

13)

$P_1 < G_1$

4) If the system has two real and equal poles, it is described as

(A) undamped (B) critically damped (C) underdamped (D) overdamped

5- A transfer function of first order system is given by:

$TF = \frac{10}{s+0.5}$, the time constant is: a) 0.5 b) 2 c) 10 d) None

4) If the system has two real and equal poles, it is described as

(A) undamped (B) critically damped (C) underdamped (D) overdamped

5- A transfer function of first order system is given by:

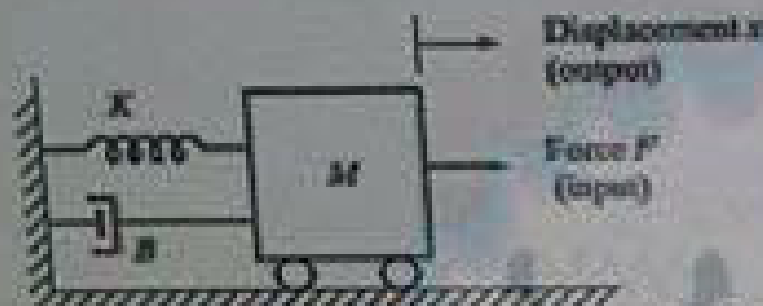
$TF = \frac{10}{s+0.5}$, the time constant is: a) 0.5 b) 2 c) 10 d) None

Q1) Select the most accurate answer

1) Choose the most accurate statement regarding the open and close loops.

- (A) Output of the open loop is dependent on reference signal
 (B) Close loop systems have less stability problems than open loops.
 (C) Open loop is easier to maintain
 (D) None

2- The system shown is of:



- (A) zero order (B) first order (C) second order (D) higher order

3) The partial fraction expansion of $Y(s) = \frac{s+2}{s^2(s+1)^2}$ is

- (A) $\frac{s+2}{s^2} + \frac{s+2}{s+1}$ (B) $\frac{s+2}{s^2} + \frac{s+2}{(s+1)^2}$ (C) $\frac{a_1}{s^2} + \frac{a_2}{s} + \frac{b_1}{(s+1)^2} + \frac{b_2}{s+1}$ (D) None

4) The inverse laplace transform of $Y(s) = \frac{1}{s^2(s+1)}$

- (A) $y(t) = t - 1 + e^{-t}$ (B) $y(t) = t^2 - t + e^{-t}$ (C) $y(t) = t^2 + t + e^{-t}$ (D) none

5) The positioning system of a printer can be modeled as

$$Y(s) = \frac{10(s+50)}{s^2 + 60s + 500} R(s).$$

where the input $R(s)$ represents the desired position and $Y(s)$ is the output position. If the input is a unit step input, the final value of the output is:

- (A) $y_{ss} = \lim_{t \rightarrow \infty} y(t) = 50$ (B) $y_{ss} = \lim_{t \rightarrow \infty} y(t) = 100$
 (C) $y_{ss} = \lim_{t \rightarrow \infty} y(t) = 1$ (D) none

6) Consider the differential equation

$$\ddot{y} + 2\dot{y} + y = u$$

Where $y(0) = \dot{y}(0) = 0$ and $u(t)$ is a unit step. The poles of this system are:
 A) $s_1 = -1, s_2 = -1$ B) $s_1 = -1, s_2 = -2$ C) $s_1 = 1j, s_2 = -1j$ D) None

Using the following matrices answer 7 and 8

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

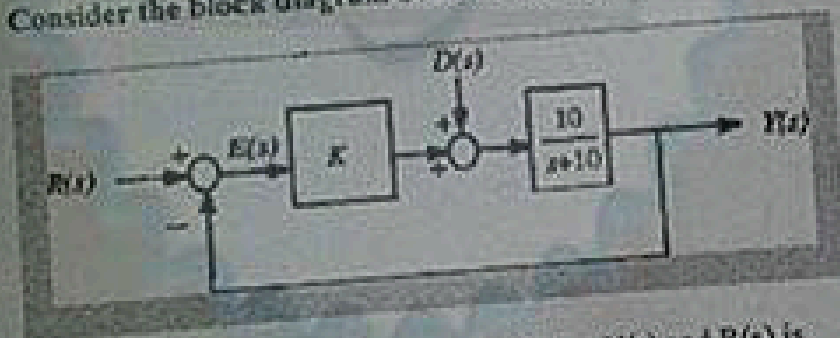
7) The state equations are

A) $\begin{cases} \dot{x}_1 = x_2 + u_2 \\ \dot{x}_2 = x_1 + u_1 \end{cases}$ B) $\begin{cases} \dot{x}_1 = x_2 - u_2 \\ \dot{x}_2 = x_1 + u_1 \end{cases}$ C) $\begin{cases} \dot{y}_1 = x_1 + u_1 \\ \dot{y}_2 = x_2 \end{cases}$ D) None

8) The output equations are

A) $\begin{cases} y_1 = x_1 + u_1 \\ y_2 = x_2 \end{cases}$ B) $\begin{cases} y_1 = x_2 + u_1 \\ y_2 = x_1 \end{cases}$ C) $\begin{cases} y_1 = x_1 + u_2 \\ y_2 = x_2 \end{cases}$ D) None

Consider the block diagram below for problems 9 and 10



9) The closed loop transfer function relating $Y(s)$ and $R(s)$ is

A) $\frac{10K}{s+10K+10}$ B) $\frac{10}{s+10K+10}$ C) $\frac{10K}{s+10}$ D) None

10) The closed loop transfer function relating $Y(s)$ and $D(s)$ is

A) $\frac{10K}{s+10K+10}$ B) $\frac{10}{s+10K+10}$ C) $\frac{10K}{s+10}$ D) None

Choose the correct answer using MATLAB.

Q1)

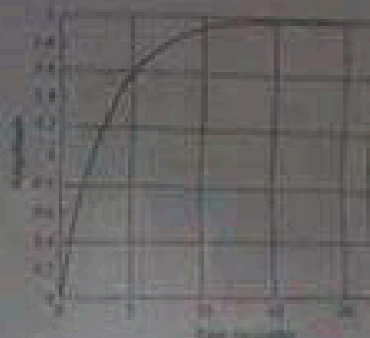
Closed-loop transfer function of a unity-feedback system is given by $T(s)/P(s) = 1/(2s + 1)$. Steady-state error to unit-ramp input is

- (A) ∞ (B) 1 (C) 1 (D) $1/2$

Q2)

The plot in Fig. shows the unit step response of a first-order system. Transfer function of the system is

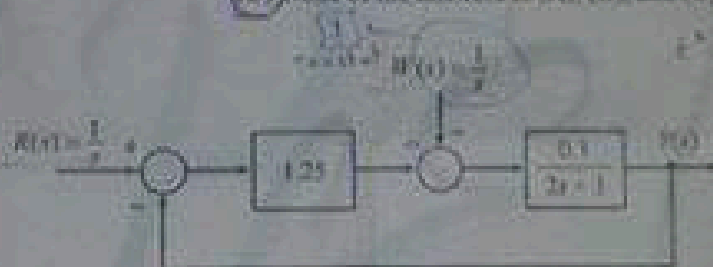
- (A) $\frac{1}{5s + 1}$ (B) $\frac{2}{15s + 1}$
(C) $\frac{2}{3s + 1}$ (D) $\frac{1}{15s + 1}$



Q3)

The system of Fig. has the steady-state error $e_{ss} = 1/2$

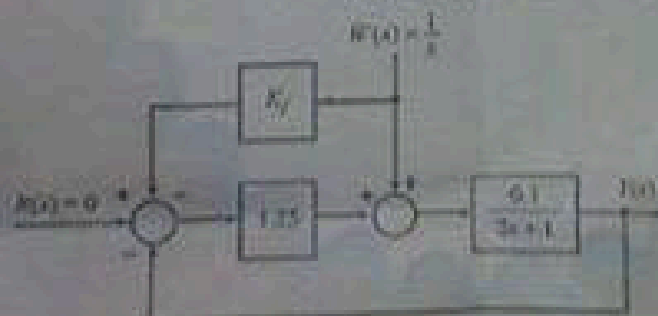
- (A) $\frac{1}{1.125}$ (B) $\frac{0.1}{1.125}$
(C) $\frac{1.1}{1.125}$ (D) None of the answers in (A), (B), and (C) is correct



Q4)

Consider the system of Fig. The steady-state error is zero for

- (A) $K_f = 0$ (B) $K_f = \frac{1}{1.25}$
(C) $K_f = 1.25$ (D) None of the answers in (A), (B), and (C) is correct



05.

$$G(x) = 20(x^2 + 5) \quad x \geq 0$$
(A) $k \geq 5$

114-5

☒ (D) All the answers in (A), (B), and (C) are correct

0204

$r(t) = K_1 + K_2 t + \frac{1}{2} t^2$. The steady-state error of the system is

eg: Input - output



Palladium

1811

100-2000

(D) None of the answers in (A), (B), and (C) is correct

071

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & c \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u$$

(A) $\frac{1}{(x+1)(x+2)}$

$$(B) \quad \frac{1}{s+1}$$
(c) $\frac{1}{x+3}$

(D) None of the answers in (A), (B), and (C) is correct

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	C	D	B	B	A	D

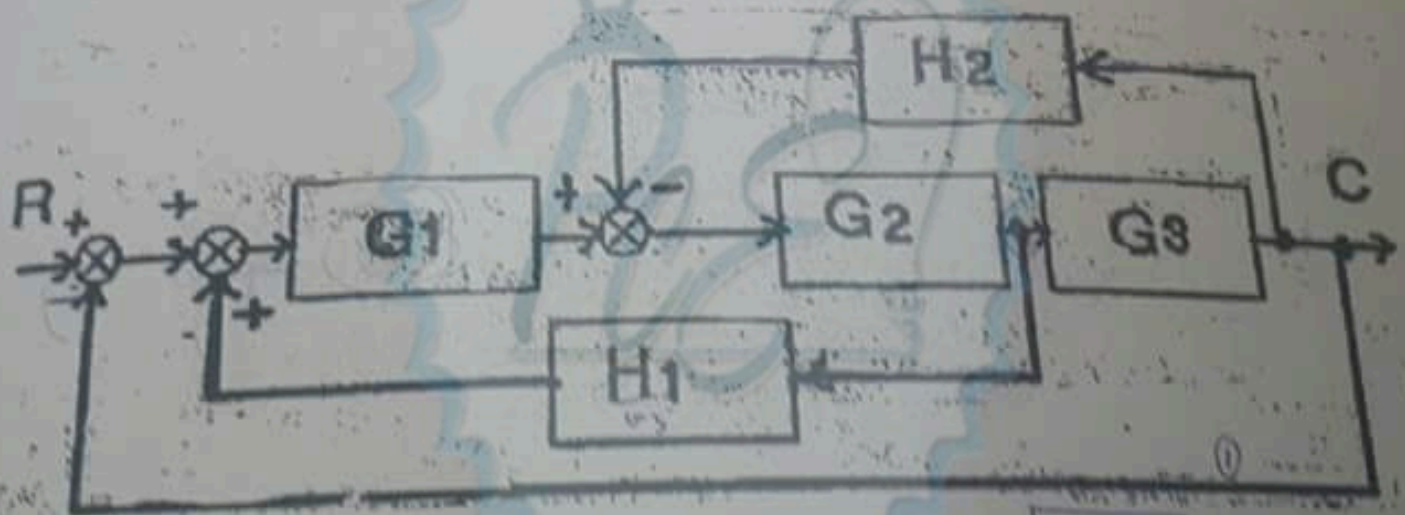
10/20

B C C B D B B

المجلد الثاني

Q2)(10 marks)

a- Reduce the following block diagram to find the transfer function of the system.





Second
Exam
20 Points

14/20

Instructors: Dr. Mahdi Al-Shamaseen
Dr. Mohammad Alkhdawaldah

Student Name: _____

Select the most correct answer from the alternatives given, and then write it in the following table

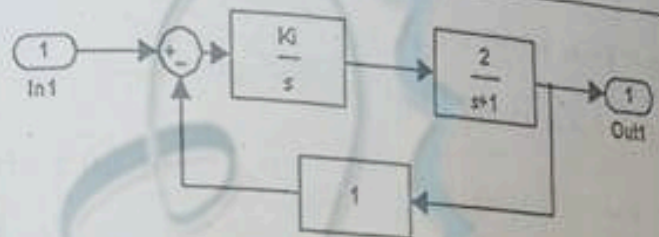
1	2	3	4	5
B	A	C	B	C

1. A unity feedback system has open-loop transfer function $G(s) = \frac{25}{s(s+6)}$, The time t_p at which the peak of the step-input response occurs is

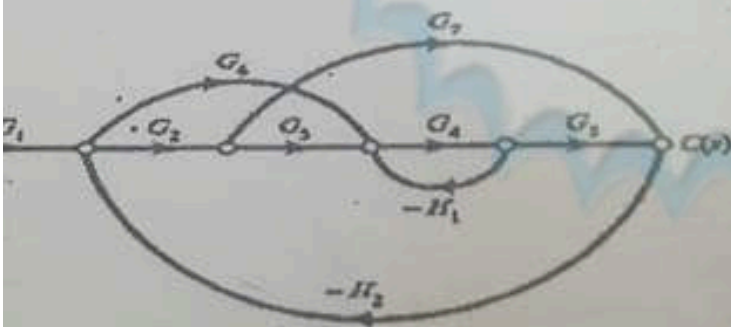
- A- 11/7 sec B- 11/4 sec C- 11/14 sec. D- 11/28 sec.

2. If $K_i=2$, the damping ratio is

A] 0.5
B] 0.25
C] 0

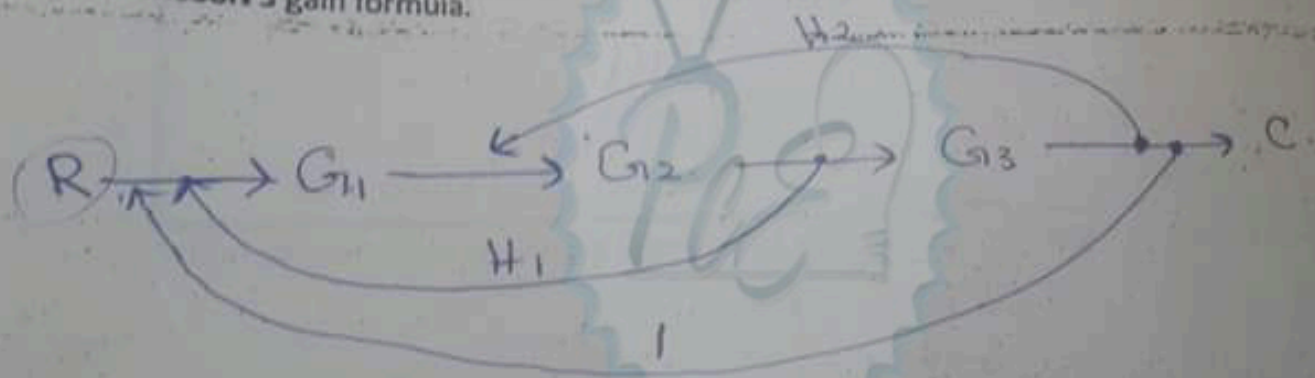


the given signal flow graph the number of closed loops is



- A] 2
B] 3
C] 4
D] 5

b- Convert the block diagram in (a) to signal flow graph and then, find the transfer function by Mason's gain formula.



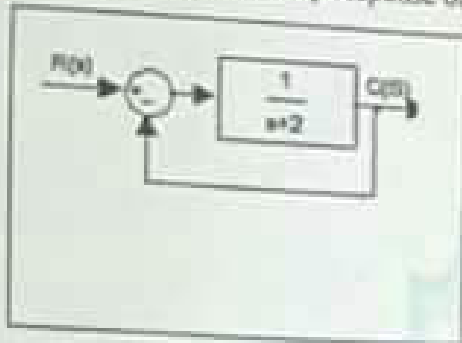
path = $G_1 G_2 G_3$

Q1 (16 marks) Select the most correct answer from the alternatives given, and then write it in the table below.

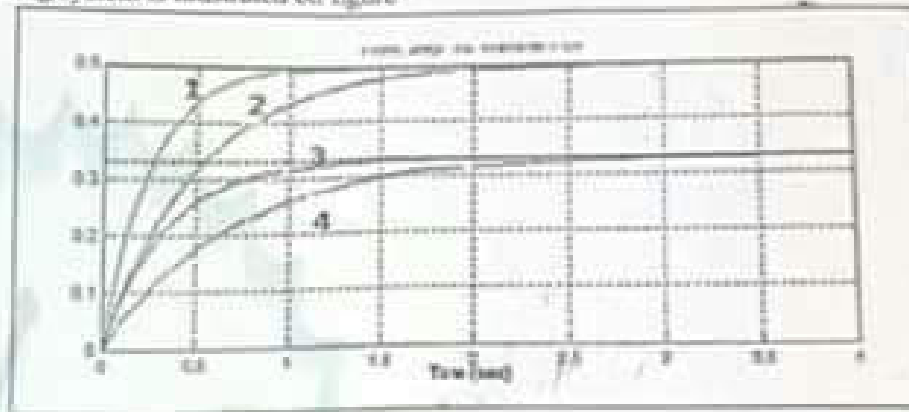
1- For most control systems the acceptable response should be
 A) over damped B) under damped C) Undamped D) critically damped

2- To eliminate the steady state error we use
 A) on-off controller B) P controller C) I controller D) D controller

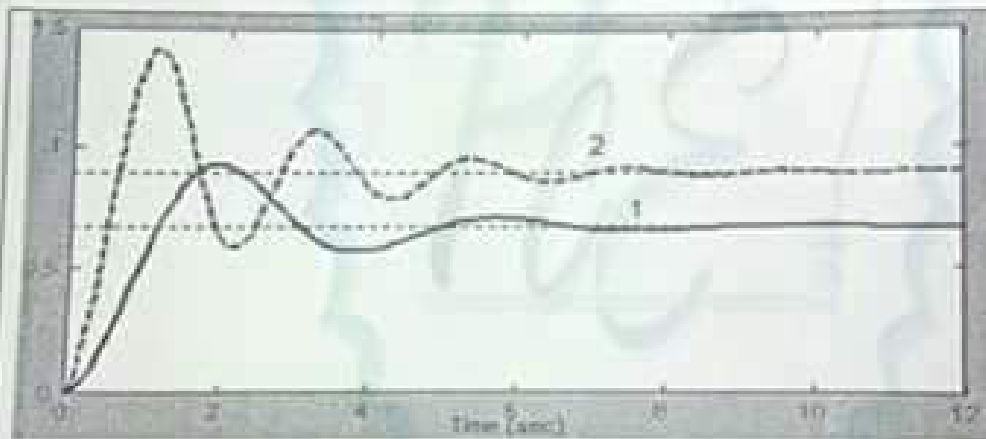
3- The unit step response of the following system is illustrated on figure



A) 4 B) 3 C) 2 D) 1

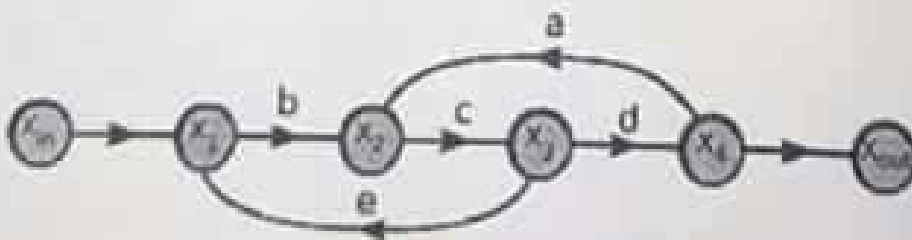


4- In the plotted figure: 1- is the unit step response of uncontrolled system, and 2- is the unit step response of controlled system. What type of controller is used?



A) P
 B) I
 C) PI
 D) PD

5- The determinant for the signal flow graph below is



- A) $1-bce-acd$ B) $1+bce+acd$ C) $1-bce-acd+bceacd$ D) $1+bce+acd-bceacd$

The characteristics equation of a closed loop control system is $s^3 + as^2 + bs + 2 = 0$. Answer 6 and 7

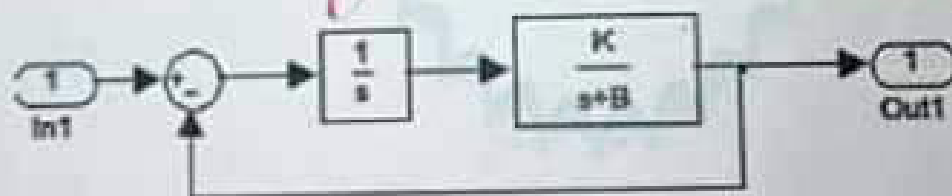
- 6- If $a=1, b=2$ the system is
 A) stable B) critically stable C) unstable D) conditionally stable
- 7- The system is stable for
 A) $a>0, b>0$ B) $a<0, b<0$ C) $ab>2$ D) $a>0, b>0, ab>2$
- 8- The mostly used controller in industry is
 A) PID controller B) on-off controller C) PI controller D) PD controller

1	2	3	4	5	6	7	8
D	C	B	D	A	A	D	C

Q2) (4 marks) For the system shown below $K=4$ and $B=2$. Determine the damping ratio and the damped frequency. Write your answer in the given space.

Damping ratio =

Damped frequency =



- 1) In steam engine speed control system the feed back element is
 A) pilot valve B) gear box C) power cylinder D) speed of rotation

- 2) The residues in partial fraction expansion of $F(s) = \frac{4s^2 + 8s + 4}{s(s+2)^3}$ is

A) 1, 2, 3 B) -1, 2, 3 C) 1, -2, 3 D) 1, 2, -3

- 3) The inverse Laplace transform of $F(s) = \frac{s+3}{s(s^2 + 5s + 10)}$ is

A) $1 - e^{-5} \cos t$ B) $1 + e^{-5} \cos t$ C) $1 - e^{-5} \cos t$ D) $e^{-5} \cos t$

- 4) in control systems feed back is an operation used to

A) fast the response B) increase the stability C) improve system dynamics
 D) reduce the difference between the output of a system and some reference input

- 5) Transfer function approach usually used when the system is

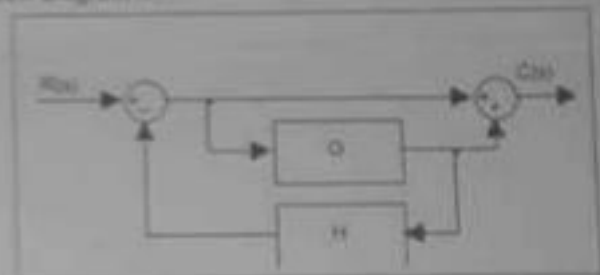
A) single input multi output B) single input single output

C) multi input single output D) multi input multi output

- 6) the transfer function from the shown block diagram is

A) $\frac{1+G}{1+GH}$ B) $\frac{1+G}{1-GH}$

C) $\frac{G}{1+GH}$ D) none



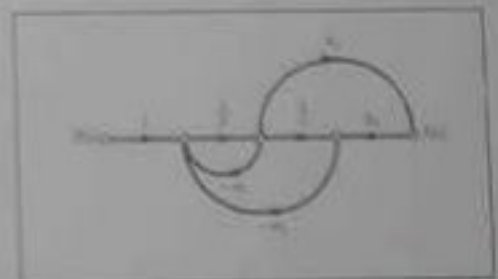
- 7) In RLC series circuit the applied voltage is the input and the voltage drop across the resistance is the output, the transfer function of this system is

A) $\frac{Cs}{LCs^2 + RCs + 1}$ B) $\frac{Ls}{LCs^2 + RCs + 1}$

C) $\frac{Rs}{LCs^2 + RCs + 1}$ D) $\frac{RCs}{LCs^2 + RCs + 1}$

- 8) the transfer function of the given signal flow graph is

A) $\frac{b_1s + b_2}{s^2 + a_1s + a_2}$ B) $\frac{b_1s + b_2}{s^2 + a_2s + a_1}$



$$C) \frac{b_1 s + b_0}{s^2 + a_1 s + a_0} \quad D) \frac{b_1 s - b_0}{s^2 + a_1 s + a_0}$$

A control system is given as

$$\dot{x}_1'(t) = x_1(t) + x_2(t) + u_1(t) + u_2(t)$$

$$\dot{x}_2'(t) = x_1(t) + u_1(t)$$

$$y_1(t) = x_1(t) + u_2(t)$$

$$y_2(t) = u_1(t) + u_2(t)$$

Answer 9 and 10

9) the state matrix **A** is

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

10) the output matrix **D** is

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$



D -

$$2) Y(s) = \frac{4s^2 + 8s + 4}{s(s+2)^2}$$

$$\frac{A}{s} + \frac{B}{(s+2)} + \frac{C}{(s+2)^2} =$$

$$A(s+2)(s+2)^2 + B(s)(s+2)^2 + Cs = 4s^2 + 8s + 4$$

$$A(s^2 + 4s + 4) + Bs^2 + 2Bs + Cs = 4s^2 + 8s + 4$$

$$As^2 + Bs^2 + 2Bs + Cs = 4s^2 + 8s + 4$$

$$As^2 + Bs^2 = 4s^2 \rightarrow A + B = 4 \rightarrow B = 3$$

$$A4s + 2Bs + Cs = 8s \rightarrow 4A + 2B + C = 8$$

$$4A = 4 \rightarrow A = 1 \quad C = 8 - 10 = -2 \quad \#C$$

$$3) Y(s) = \frac{s+3}{s(s^2+6s+10)} \rightarrow \frac{A}{s} + \frac{Bs+C}{s^2+6s+10}$$

$$A(s^2+6s+10) + B(s+C)s = s+3$$

$$A = 0.3 \rightarrow A+B=0$$

$$6A+C=1$$

$$10A=3 \Rightarrow B = -0.3, C = -0.8$$

$$\frac{0.3}{s} + \frac{-(0.3s+0.8)}{s^2+6s+10+9-9}$$

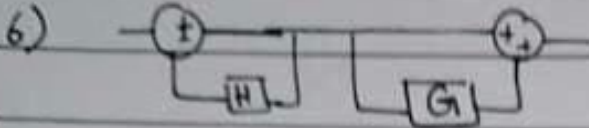
$$\frac{0.3}{s} + \frac{-(0.3s+0.8) + 0.1 - 0.1}{(s+3)^2 + 1}$$

$$0.3 - 0.3(\cos t \cdot e^{-3t} - 0.1 \sin t \cdot e^{-3t})$$

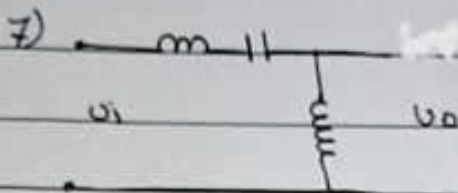


4) D

5) B



$$\frac{1}{1+HG} \cdot (1+G) \quad \#A$$



$$u_1 = L \frac{di}{dt} + \frac{1}{C} \int i dt + R i(A)$$

$$u_2(s) = L(s) + \frac{1}{Cs} I(s) + R I(s)$$

$$\Rightarrow \frac{I(s)}{u_2(s)} = \frac{1}{L(s) + \frac{1}{Cs} + R} \Rightarrow \frac{Cs}{CsLs + 1 + CSR}$$

$$\frac{u_0}{s} = R I s \rightarrow \frac{u_0}{I s} = R$$

$$\frac{u_0}{u_2} = \frac{RCS}{CsLs + 1 + CSR} \quad \#D$$

8) $P_1 =$

$$1 \quad \frac{1}{s} \quad \frac{1}{s} \quad b_2$$

$P_2 =$

$$1 \quad \frac{1}{s} \quad b_1$$

9) #B

10) #D

$$L_1 = -a_1 \frac{1}{s}, \quad L_2 = -a_2 \frac{1}{s} \frac{1}{s}$$

$$\Delta = 1 - (-a_1 \frac{1}{s} + -a_2 \frac{1}{s} \frac{1}{s}) + 0$$

$$\Delta = 1 + a_1 \frac{1}{s} + a_2 \frac{1}{s} \frac{1}{s}$$

$$\Delta_1 = 1$$

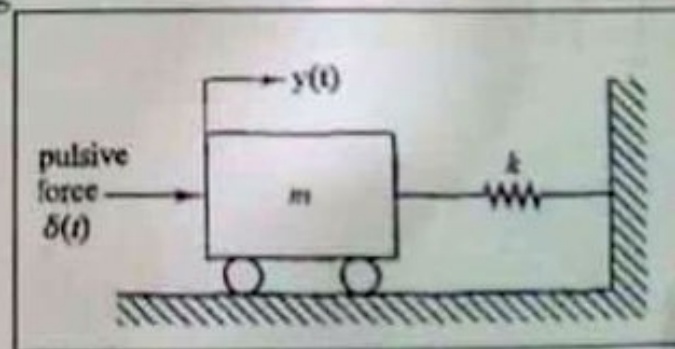
$$\Delta_2 = 1$$

$$t.f. = \frac{P_1 \Delta_1 + \Delta_2 P_2}{\Delta} = \frac{1 \cdot 1 + 1 \cdot b_1}{1 + a_1 \frac{1}{s} + a_2 \frac{1}{s}}$$

$$= \frac{(\frac{1}{s} b_2 + b_1 \frac{1}{s})}{(1 + a_1 \frac{1}{s} + a_2 \frac{1}{s})} s^e = \frac{b_2 + b_1 s}{s^2 + a_1 s + a_2} \quad \#A$$

1	2	3	4	5	6	7	8	9	10	11	12	13	14
B	A	D	D	C	A	B	A	A	B	A	C	D	D

1. Let $F(s) = \frac{1}{s(s+1)}$, The final value of $f(t)$ is: $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} \frac{1}{s+1} = 1$
 a) 0 ☒ b) ∞ c) 1 d) None
2. $y(t) = e^{-t} - e^{-2t}$, Then $Y(s)$ will be
☒ A. $\frac{1}{s^2+1}$ B. $\frac{-2}{s^2-1}$ C. $\frac{+2}{s^2-1}$ D. None
3. Open-loop control can be used, in practice, only if:
 A. the relationship between the input and output is known B. there are no internal disturbances.
 C. there are no external disturbances ☒ D. all above
4. The transfer function of an L C series circuit, where $v_c(t)$ is the output and the applied voltage $v(t)$ is the input, is
 A. $\frac{L}{LCs+1}$ ☒ B. $\frac{1}{LCs^2+1}$ C. $\frac{Cs}{LCs^2+1}$ ☒ D. $\frac{1}{LCs+1}$
5. The number of state equations in spring-mass-dashpot linear mechanical system is
 A. 4 B. 3 ☒ C. 2 D. 1
6. The output of the main comparator in a feedback control system is called
☒ A. error signal B. Feedback signal C. output signal. D. Difference signal
7. The major disadvantage of a feedback system may be
 A. Inaccuracy ☒ B. instability C. Unreliability D. inefficiency
8. For the system shown, the mathematical model relating The output ($y(t)$) and the input ($\delta(t)$) is
☒ A. $my''(t) + fy'(t) + Ky(t) = \delta(t)$
 B. $my''(t) + Ky(t) = \delta(t)$
 C. $my''(t) + fy'(t) + Ky(t) = 0$
 D. $my''(t) + Ky(t) = 0$
9. The state variables of a system given by $y'''(t) + 5y''(t) + 4y'(t) + 2y(t) = 3u(t)$ are
☒ A. $x_1(t) = y(t); x_2(t) = y'(t); x_3(t) = y''(t)$ B. $x_1(t) = 2y(t); x_2(t) = 4y'(t); x_3(t) = 5y''(t)$
 C. $x_1(t) = y(t); x_2(t) = r(t)$ D. $x_1(t) = y(t); x_2(t) = y''(t)$
10. The steady state error in the unit ramp response of first order standard system is
 A. 0 ☒ B. T C. 2T D. 3T



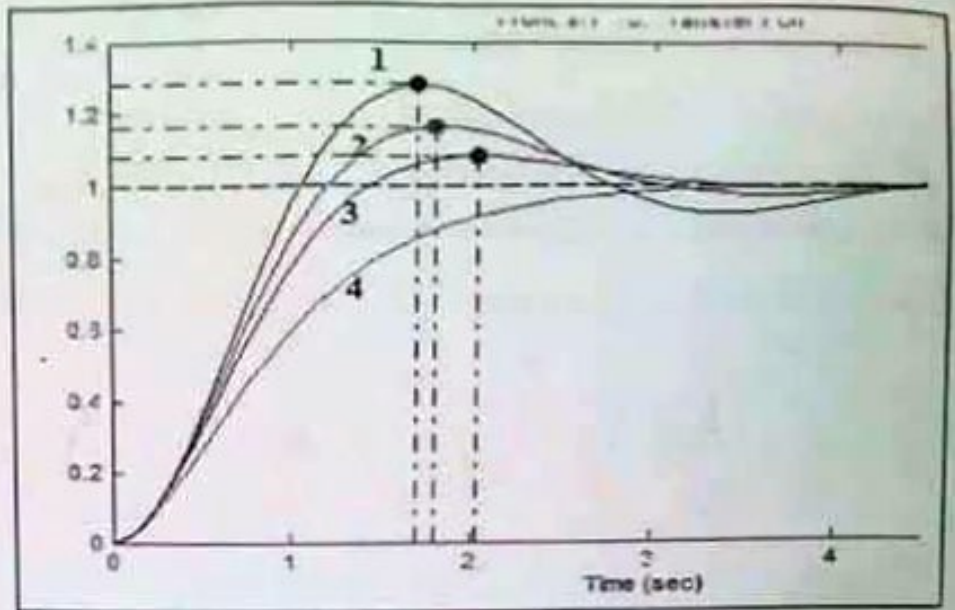
Depend on the following figure to solve 11 & 12.

11. The time response of a second Order system given by

$$\frac{C(s)}{R(s)} = \frac{4}{s^2 + 2s + 4}$$

is denoted as

- ☒ A. 1 B. 2 ☐ C. 3 D. 4



12. The unacceptable response with respect to the rise time is denoted by

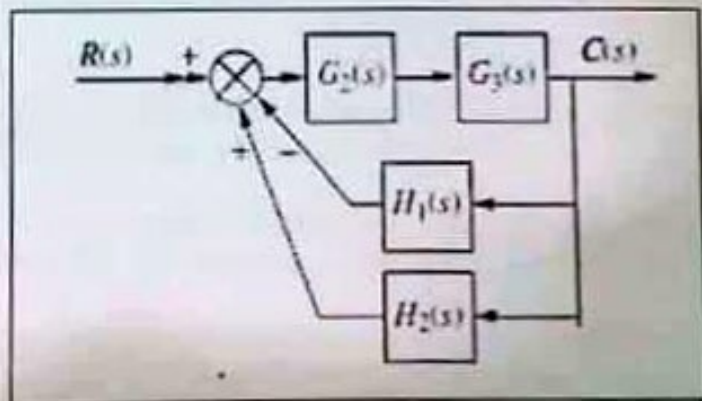
- ☒ A. 1 B. 2 C. 3 D. 4

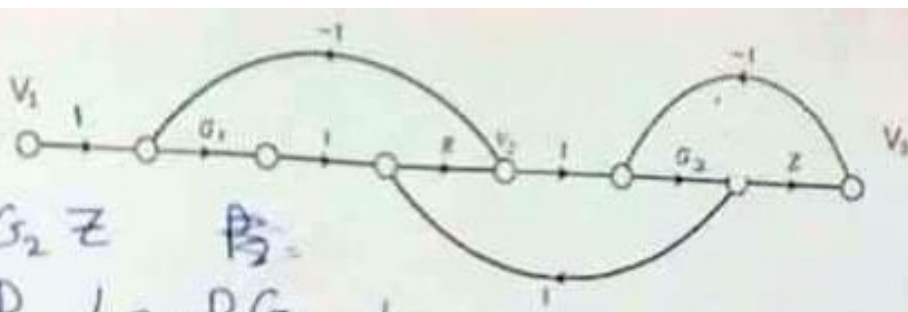
13. The partial fraction of $y(s) = \frac{1}{s^2(s+1)^2}$ is

- A. $y(s) = \frac{1}{s^2(s+1)^2}$ B. 2 $y(s) = \frac{a}{(s+1)^2} + \frac{b}{s^2}$
☒ C. $y(s) = \frac{\alpha_1 s + \alpha_2}{(s+1)^2} + \frac{b}{s^2}$ ☐ D. $y(s) = \frac{\alpha_1 s + \alpha_2}{(s+1)^2} + \frac{\beta_1 s + \beta_2}{s^2}$ ← ans D

14. The open loop transfer function from the given block diagram is

- A. $G_2 G_3 H_1 H_2(s)$
 B. $G_2 G_3 H_1(s)$
 C. $G_2 G_3 H_2(s)$
☒ D. $G_2 G_3 (H_2(s) - H_1(s))$





$$P_1 = G_1 R G_2 Z$$

$$L_1 = -G_1 R \quad L_2 = R G_2 \quad L_3 = -G_2 Z$$

untouch loop

$$L_1 L_2 = -G_1 R^2 \quad L_1 L_3 = +G_1 G_2 R Z \quad L_2 L_3 = -R G_2^2 Z$$

$$\Delta = 1 - [-G_1 R + R G_2 - G_2 Z] + [-G_1 R^2 + G_1 G_2 R Z - R G_2^2 Z]$$

$$\Delta_1 = 1$$

$$\frac{V_3}{V_1} = \frac{\Delta_1 \times P_1}{\Delta} = \frac{1 \times G_1 R G_2 Z}{1 - [-G_1 R + R G_2 - G_2 Z] + [-G_1 R^2 + G_1 G_2 R Z - R G_2^2 Z]}$$

Q3) (6 marks) Assume zero initial conditions, solve the following differential equation using Laplace transform

$$y''(t) + 3y'(t) + 2y(t) = 1$$

$$s^2 y(s) + 3s y(s) + 2y(s) = \frac{1}{s}$$

$$y(s) [s^2 + 3s + 2] = \frac{1}{s}$$

$$y(s) = \frac{1}{s[s^2 + 3s + 2]} \Rightarrow \frac{1}{s(s+2)(s+1)} \Rightarrow \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+1}$$

$$A(s+2)(s+1) + B(s+1)s + C(s+2)s = 1$$

$$A = \frac{1}{2} \quad B = \frac{1}{2} \quad C = -1$$

$$\Rightarrow \frac{1}{2} + \frac{1}{2} e^{-2t} - e^{-t}$$



$$1) F(s) = \frac{1}{s(s+1)} \Rightarrow \lim_{s \rightarrow 0} F(s) = \lim_{s \rightarrow 0} s \frac{1}{s(s+1)} = 1 \quad \# \textcircled{C}$$

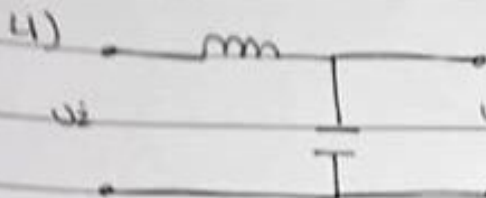
$$\text{OR } F(s) = \frac{A}{s} + \frac{B}{s+1} \rightarrow A(s+1) + B(s) = 1, A=1, B=-1$$

$$F(s) = \frac{1}{s} + \frac{-1}{s+1} \quad F(t) = 1 - e^{-t}$$

$$\lim_{s \rightarrow \infty} F(s) = 1 \quad \# \textcircled{C}$$

$$2) Y(t) = \frac{-t}{e} e^t \sim Y(t) = -2 \sinh t \rightarrow Y(s) = \frac{-2}{s^2+1} \quad \# \textcircled{B}$$

$$3) \# \textcircled{D}$$



$$u_i = L \frac{di}{dt} + \frac{1}{C} \int I(t) dt$$

$$u_i(s) = L(s) I(s) + \frac{1}{Cs} I(s)$$

$$\frac{u_i(s)}{u_o(s)} = \frac{1}{2s + \frac{1}{Cs}} \rightarrow \frac{Cs}{Cs^2 + 1}$$

$$\Rightarrow u_o = \frac{1}{C} \int I(t) dt \Rightarrow u_o(s) = \frac{1}{Cs} I(s)$$

$$\frac{u_o(s)}{u_i(s)} = \frac{1}{Cs^2 + 1} = \frac{1}{Cs^2 + 1} \quad \# \textcircled{B}$$

$$5) \# \textcircled{C}$$

$$6) \# \textcircled{A}$$

$$7) \# \textcircled{B}$$

$$8) m \ddot{y}(t) + k \dot{y}(t) = \delta(t) \quad \# \textcircled{B}$$

$$F = k \dot{y}(t)$$

الذنب

$$a) \ddot{y}(t) + 5\dot{y}(t) + 4y(t) + 2y(t) = 3u(t)$$

$$x_1 = y(t) = x_2$$

$$x_2' = \dot{y}(t) = x_3$$

$$x_3' = \ddot{y}(t) = -5\dot{y}(t) - 4y(t) - 2y(t) + 3u(t) \quad \# \textcircled{A}$$



Power and Communication Engineering

10) $r(t) = t$ ✗ (B)

11) ✗ B $\rho_{\text{avg}} = \frac{1}{T} \int_0^T \rho(t) dt$

~~steady state~~ $t_p = \frac{\pi}{\omega_d} = 1.8 \approx 2$

12) ✗ (D) H_{zero}

13) $\frac{1}{s^2(s+1)^2} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1} + \frac{D}{s+1} \quad \text{✗ (D)}$

14) ✗ D Feedback $\rightarrow$ open loop

Q2) $P_1 = 1 \quad G_1 \quad R \quad G_2 \quad Z$

$L_1 = G_1 R - 1$

$L_2 = R G_2 (1)$

$L_3 = G_2 Z - 1$

non tech loop

$L_4 L_3 = -G_1 R - G_2 Z$

$\Delta = 1 - \sum \text{loop} + \sum \text{non tech}$

$= 1 - (-G_1 R - G_2 Z + R G_2) + G_1 G_2 R Z$

$= 1 + G_1 R + G_2 Z - R G_2 + G_1 G_2 R Z$

$\Delta_1 = 1$

$\frac{V_o}{V_i} = \frac{\Delta R_1}{\Delta} = \frac{1 G_1 G_2 R Z}{1 + G_1 R + G_2 Z - R G_2 + G_1 G_2 R Z}$

Q3) $y''(t) + 3y'(t) + 2y(t) = 1$

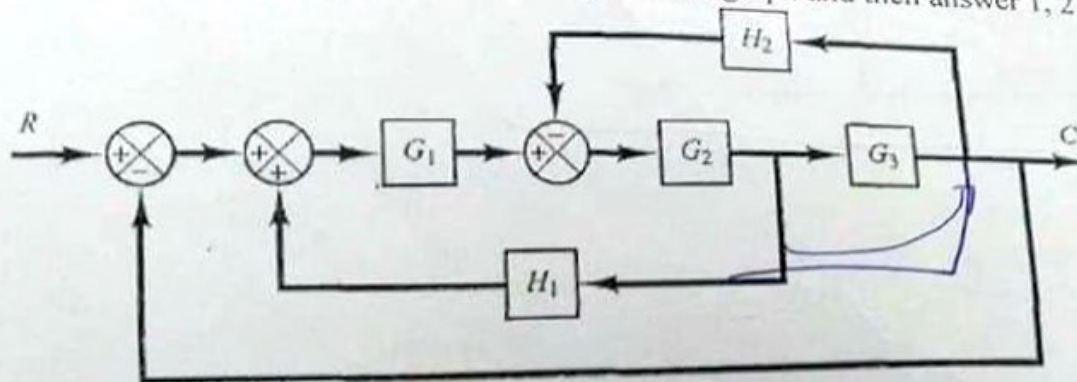
Sol: $s^2 y(s) + 3s y(s) + 2y(s) = \frac{1}{s}$

$y(s) = \frac{1}{s} \cdot \frac{1}{s^2 + 3s + 2} = \frac{1}{s} \cdot \frac{1}{(s+2)(s+1)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+1}$

$A(s+2)(s+1) + B s(s+1) + C (s+2)(s) = 1 \quad | A = \frac{1}{2}, B = \frac{1}{2}, C = -1$

$y(s) = \frac{\frac{1}{2}}{s} + \frac{\frac{1}{2}}{s+2} + \frac{-1}{s+1}$
 $y(t) = \frac{1}{2} + \frac{1}{2} e^{-2t} - e^{-t}$

Convert the block diagram below to a signal flow graph and then answer 1, 2 & 3. (35 marks)



1- The number of non-touching loops (two at a time) is:
a) 2 b) 1 **c) 0** d) None

2- The determinant is equal:
a) $1 - G_1 G_2 H_1 - G_2 G_3 H_2 + G_1 G_2 G_3$ **b) $1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3$** c) 1 d) None

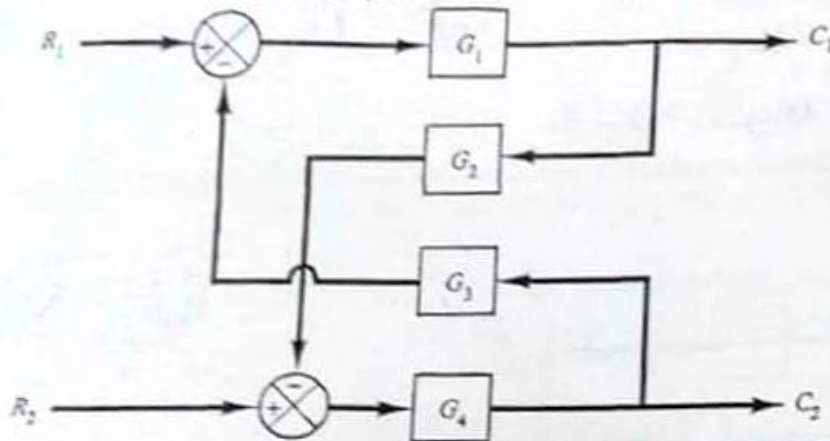
3- The transfer function $\frac{C(s)}{R(s)}$ is:

a) $\frac{G_1 G_2 G_3}{1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3}$ b) $\frac{G_1 G_2 H_2}{1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3}$ c) $\frac{G_1 G_2 G_3}{1 - G_1 G_2 H_1 - G_2 G_3 H_2 + G_1 G_2 G_3}$ d) None

4- The determinant of the forward path $G_1 G_2 G_3$ is:
a) 1 b) $1 - H_1 H_2$ c) $1 + H_1 H_2$ d) None

5- The figure shown below represents a linear time invariant system, the transfer function C_1/R_2 of this system is

- a) $-\frac{G_1 G_3 G_4}{1 - G_1 G_2 G_3 G_4}$ b) $-\frac{G_1 G_3 G_4}{1 + G_1 G_2 G_3 G_4}$ c) $\frac{G_1 G_3 G_4}{1 + G_1 G_2 G_3 G_4}$ d) None



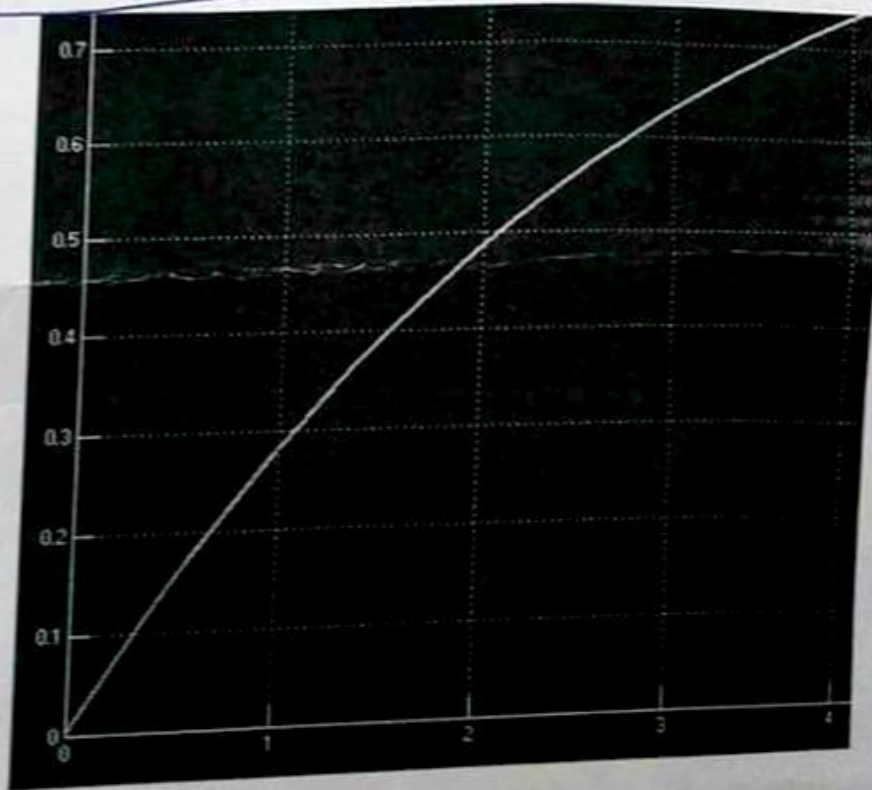
The adjacent figure shows the unit step time response of first order system. Use this figure to answer 6 & 7.

6- The time constant is

- a) 1 sec b) 2 sec. c) 3 sec. d) 4 sec

7- The settling time is:

- a) 6 sec b) 8 sec c) 17 sec d) 12 sec



8- One of the following is not true for Open loop control system

- a) Less expensive than close loop systems. b) less stable than close loop systems
c) Simpler to construct than close loop. d) None

9- The function $F(s) = \frac{s+4}{s^2(s+2)^2(s+5)}$ can be expanded with partial fraction as follows:

a) $\frac{a}{s} + \frac{a}{s^2} + \frac{b}{s+2} + \frac{b}{(s+2)^2} + \frac{c}{s+5}$ b) $\frac{a}{s} + \frac{b}{s^2} + \frac{c}{s+2} + \frac{d}{(s+2)^2} + \frac{e}{s+5}$

~~c) $\frac{as+b}{s^2} + \frac{as+b}{(s+2)^2} + \frac{as+b}{(s+5)}$~~ d) None

10- The inverse Laplace transforms of $F(s) = \frac{2s+3}{s^2+4s+13}$ is: A) $F(s) = e^{-2t}(2\cos 3t - \sin 3t)$
B) $f(t) = e^{-2t}(2\cos 3t - \frac{1}{3}\sin 3t)$ C) $f(t) = e^{-2t}(2\cos 3t + \frac{1}{3}\sin 3t)$ D) None.

11- The initial value of $F(s) = \frac{2}{s(s+1)}$ is:

- a) 1 b) 2 c) 0 d) 4

For an RC series circuit, the input voltage is $e_i(t)$ and the output is the voltage across the capacitor. Use this information to answer 12, 13 & 14.

12- The Transfer function is:

a) $\frac{1}{RCS+1}$ b) $\frac{CS}{RCS+1}$ c) $\frac{R}{RCS+1}$ d) None

$$RC \frac{di}{dt} + \frac{1}{C} \int i dt = e_i$$

$$e_o = \frac{1}{C} \int i dt$$

13- The differential equation that describes the system is:

a) $e'_o(t) = \frac{e_i(t) + e_o(t)}{RC}$ b) $e'_o(t) = \frac{e_i(t) + e'_o(t)}{RC}$ c) $e'_o(t) = \frac{e_i(t) - e_o(t)}{RC}$ d) None

14- For state space representation, the matrix A =

a) $[\frac{1}{RC}]$ b) $[-\frac{1}{RC}]$ c) $[-1]$ d) None

1	2	3	4	5	6	7	8	9	10	11	12	13	14
c	b	a	d	c	c	a	d	d	A	C	D	C	b

22.5

Q2) (5 marks) Solve the following differential equation: $X''(t) + 2X'(t) + 10X(t) = 0$
 $X'(0) = 0, X(0) = 0$



1) 0 $\neq$ C

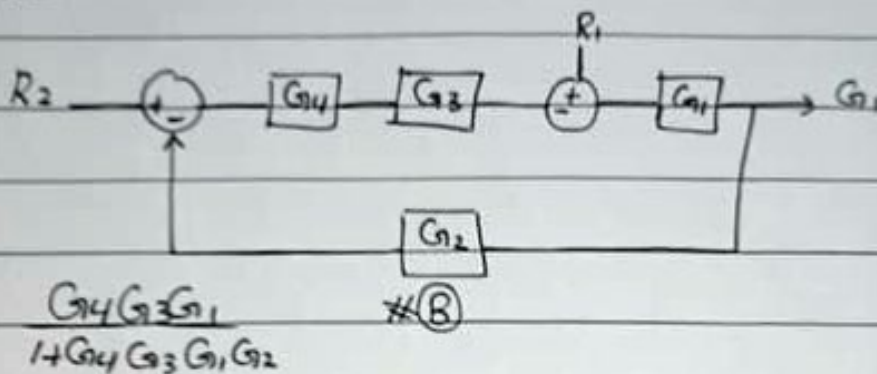
2) $\Delta = 1 - \sum \text{Loops} + \sum \text{non touching}$

$\Delta = 1 - G_1 G_2 H_2 + G_2 G_1 H_1 + G_1 G_2 G_3$ $\neq$ b

3) $\frac{G_1 G_2 G_3}{1 - G_1 G_2 H_1 + G_2 G_1 H_2 + G_1 G_2 G_3}$ $\neq$ a

4) $\Delta_{G_1 G_2 G_3} = 1$ $\neq$ a

5) $\frac{G_1}{R_2}$



6) $\gamma = 0.63$ steady state = 0.68

$\tau = 3 \text{ sec}$ $\neq$ C

7) 12 sec $\neq$ D $\rightarrow T_s = 4$ $T_{\text{constant}} = 12 \text{ sec}$

8) $\neq$ B

9) $F(s) = \frac{s+4}{s^2(s+2)^2(s+5)}$

$= \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+2} + \frac{D}{(s+2)^2} + \frac{E}{s+5}$ $\neq$ B

10) $F(s) = \frac{2s+3}{s^2+4s+13}$

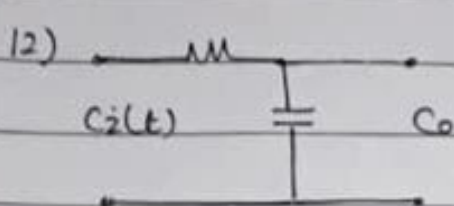
Sol: $\frac{2s+3}{s^2+4s+13} = \frac{2s+3}{s^2+4s+13.4-4} = \frac{2s+3}{s^2+4s+9+4}$



$$= \frac{2s+3}{(s+2)^2+9} = \frac{2s+3+1-1}{(s+2)^2+9} = \frac{2(s+2)-1}{(s+2)^2+9}$$

$$= \frac{2(s+2)}{(s+2)^2+9} - \frac{1}{(s+2)^2+9} = 2 \cos 3t \cdot e^{-2t} - \frac{1}{3} \sin 3t \cdot e^{-2t} \quad \# B$$

11) *C



$$c_i(t) = R I(t) + \frac{1}{C} \int I(t) dt$$

$$C_i(s) = R I(s) + \frac{1}{Cs} I(s)$$

$$C_i(s) = I(s) \left(R + \frac{1}{Cs} \right) \rightarrow \frac{I(s)}{C_i(s)} = \frac{1}{CsR+1}$$

$$\frac{I(s)}{C_i(s)} = \frac{C(s)}{CsR+1}$$

$$C_o(t) = \frac{1}{C} \int I(t) dt \rightarrow C_o(s) = \frac{1}{Cs} I(s)$$

$$\frac{C_o(s)}{I_o(s)} = \frac{1}{Cs} \Rightarrow \frac{C_o(s)}{C_i(s)} = \frac{1}{CsR+1} \quad \# A$$

13) $C_o = \frac{1}{C} \int I(t) dt \Rightarrow C_o' = \frac{1}{C} I(t)$

$$I(t) = \frac{c_i - c_o}{R} \Rightarrow C_o = \frac{C_i - C_o}{Rc} \quad \# C$$

14) $C_i(t) = R I(t) + \frac{1}{C} \int I(t) dt$

$$x_1 = \int I(t) dt$$

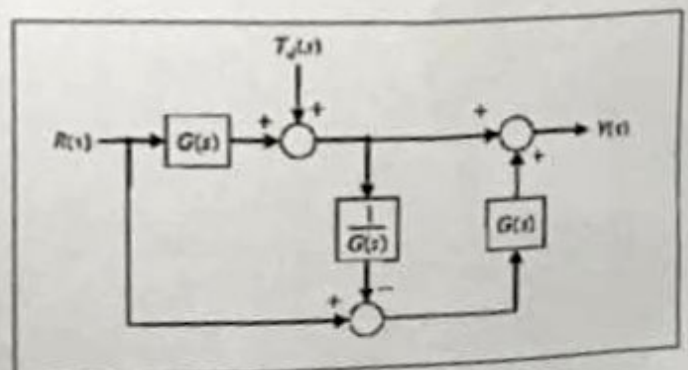
$$x_1' = I(t) = \frac{C_i(t)}{R} - \frac{1}{CR} \int I(t) dt$$

$$[x_1'] = \overset{A}{\begin{bmatrix} -1 \\ CR \end{bmatrix}} [x_1] + \overset{B}{\begin{bmatrix} 1 \\ R \end{bmatrix}} [C_i]$$

$$\# A \begin{bmatrix} -1 \\ CR \end{bmatrix} \quad \# B$$

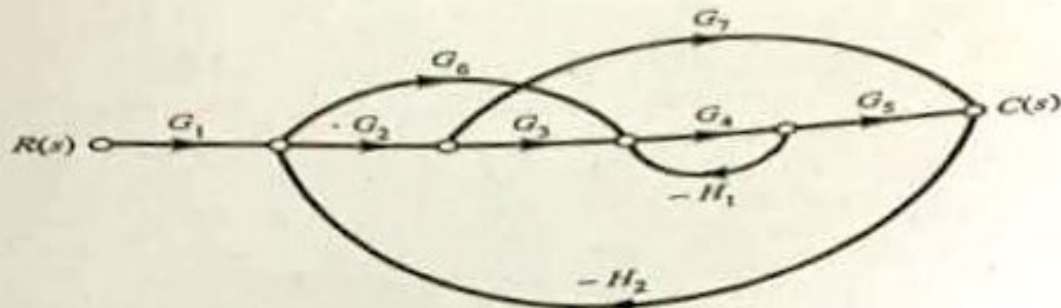
1. The system given by $m(t)y''(t) + ky(t) = u(t)$ is
☒ A. Nonlinear B. time invariant C. Time varying ☒ D. linear
2. The transfer function of an L C series circuit, where $v_c(t)$ is the output and the applied voltage $v(t)$ is the input, is
 A. $\frac{1}{LCs^2 + 1}$ ☒ B. $\frac{Cs}{LCs^2 + 1}$ C. $\frac{1}{LCs + 1}$ D. $\frac{L}{LCs + 1}$
3. The state space model of an translation mechanical system, where displacement is the output and the applied force is the input, includes an input matrix $B(s)$ with
☒ A. 1×1 B. 2×1 C. 1×2 D. 2×2
4. The inverse Laplace transforms of $Y(s) = \frac{10(s+4)}{s^2 + 8s + 15}$ is
 A. $y(t) = 10e^{-5t} + 10e^{-3t}$ B. $y(t) = 10e^{3t} + 10e^{-3t}$
 C. $y(t) = 10e^{3t} - 10e^{-3t}$ ☒ D. none
5. one of the following is an advantage of closed loop system
 A. low coast B. easy to maintain C. always stable ☒ D. high accuracy
6. "feed back" is operation used to:
 A. measure the output signal B. generate feed back signal
☒ C. reduce the difference between the output and reference input
 D. determine the feed back block transfer function
7. In speed control system the feedback element is
☒ A. pilot valve B. control valve
☒ C. gear box D. power cylinder

8. The transfer function relating $Y(s)/R(s)$ is



A] $-G(s)$	B] $1+G(s)$
C] $G(s)$	D] $G(s) + T_d(s)$

9. The number of nontouching loops in the signal flow graph is



A] 0	B] 2
C] 3	D] 4

A first-order system is represented by the differential equation $y'(t) + 4y(t) = 2u(t)$

Answer 10 and 11

10. the time constant of this system is

A. 2 B. 4 C. 0 D. 0.25

11- the steady state value of the output is

A. 0 B. 1 C. 2 D. 0.5

For a second order system given as $\frac{C(s)}{R(s)} = \frac{1}{s^2 + 5s + 4}$

answer 12 and 13

12 the unit step response of this system is

A. over damped B. under damped C. undamped D. critically damped

13. the damping ratio is

A. 0.8 B. 1 C. 0.25 D. 2



(Institution X)

1) One of the following is an advantage of closed loop.

- A) Low cost B) easy to maintain C) always stable
D) high accuracy.

2) The system given by $m(t) \ddot{y}(t) + ky(t) = u(t)$ is

- A) Non linear **B) Time varying** C) Time invariant D) Linear

3) The Transfer Function of Ls series circuit where $I(t)$ is the output and $V(t)$ is the input.

- A) $\frac{1}{Ls^2+1}$ **B) $\frac{Cs}{Ls^2+1}$** C) $\frac{1}{Ls+1}$ D) $\frac{1}{Ls+1}$

4) The state space model of an transfer mechanical system, where displacement is the output and the applied force is input. $BC(s)$ with.

- A) 1×1 B) 1×2 **C) 2×1** D) 2×2

5) The inverse laplace $Y(s) = \frac{10(s+1)}{s^2+8s+15}$ is

- A) $Y(t) = 10e^{-5t} + 10e^{-4t}$ B) $Y(t) = 10e^{+5t} + 10e^{3t}$
C) $Y(t) = 10e^{5t} - 10e^{3t}$ **D) None**

6) Feed back is used As :-

- A) measure the output signal B) generate Feed back signal
C) determine the Feed back block Transfer Function **D) reduce the different between output and Input.**

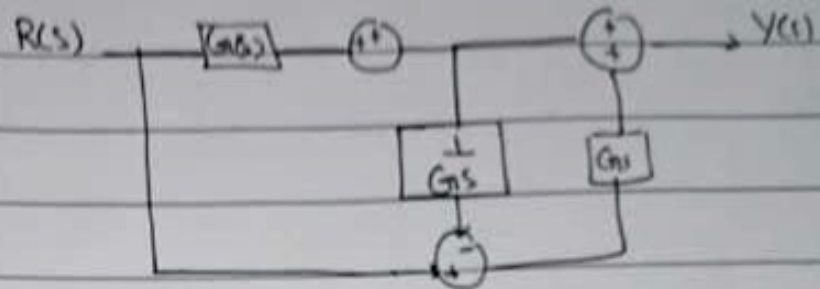


7) In speed control system the Feed back is

- A) pilot value B) control value
C) power cylinder **D) gear Box**

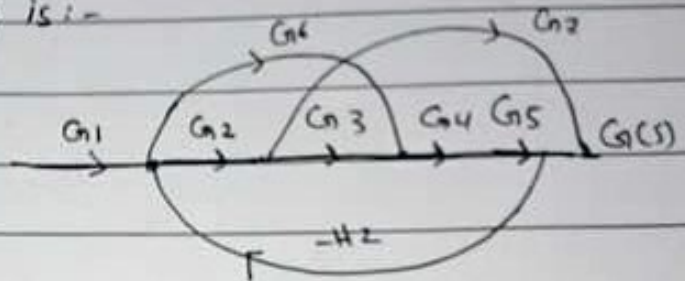
8) $X(s) / R(s)$

$G_1(s)$



9) number of non touching loop is:-

- A) 0 B) 4
C) 2 D) 2



10) $2\ddot{y}(t) + 4\dot{y}(t) = 24$

The Time constant is

- A) 2 B) 4 C) 0 **D) 0.5**

11) The steady state value is:

- A) 0 B) 1 C) 2 **D) 0.5**

12) For $\frac{C(s)}{R(s)} = \frac{1}{s^2 + 5s + 4}$, The unit step response is:-

- A) over damped B) under damped C) undamped
D) critical damped

13) The damping ratio is

- A) 1 **B) 0.25** C) 2 D) 4



Q2) $\ddot{x}(t) + 2\dot{x}(t) + 10x(t) = e^{-t}$

$\dot{x}(0) = 0, x(0) = 0$

Sol: $s^2 X(s) + 2sX(s) + 10X(s) = \frac{1}{s+1}$

$X(s)(s^2 + 2s + 10) = \frac{1}{s+1}$

$X(s) = \frac{1}{(s+1)(s^2 + 2s + 10)} = \frac{A}{s+1} + \frac{Bs+C}{s^2 + 2s + 10}$

$A(s^2 + 2s + 10) + Bs^2 + Bs + Cs + C = 1$

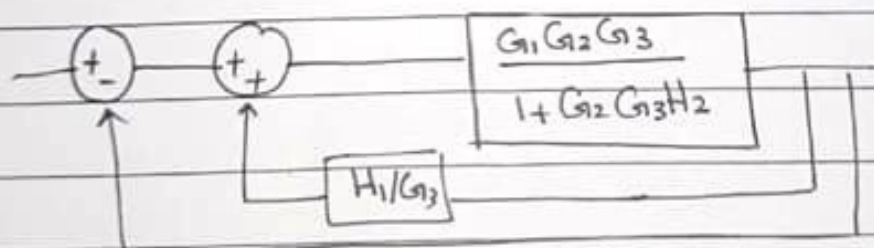
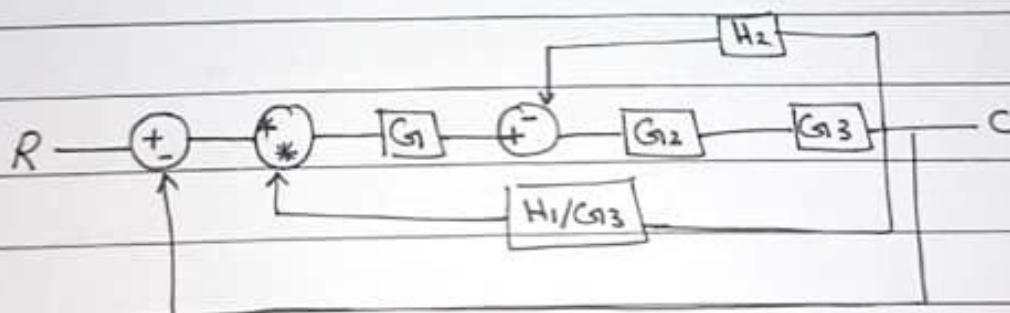
$As^2 + Bs^2 = 1 \rightarrow A + B = 0$ s^2 ko baad

$2A + Bs + Cs = 0 \rightarrow 0$ s ko baad

$10 + C = 1 \Rightarrow A = \frac{1}{9}, B = -\frac{1}{9}, C = -\frac{1}{9}$

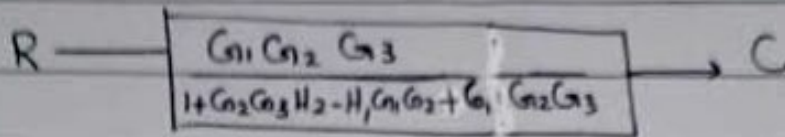
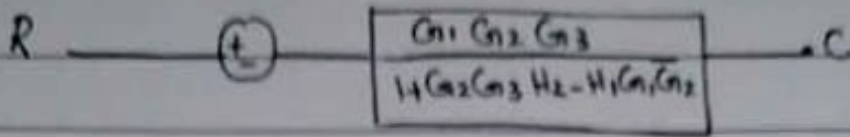
$= \frac{1}{9} + \frac{-\frac{1}{9}s - \frac{1}{9}}{s^2 + 2s + 10 + 1} = \frac{1}{9} e^{-t} + \frac{-\frac{1}{9}(s+1)}{(s+1)^2 + 9}$

$= \frac{1}{9} e^{-t} - \frac{1}{9} \cos 3t \cdot e^{-t}$ *





Power and Communication Engineering



Al-Balqa' Applied University



Faculty of Engineering Technology
Mechatronics Engineering

Midterm Exam- First Semester 2018-2019 Control Theory

Duration: : 80 minutes only

Student Name, _____

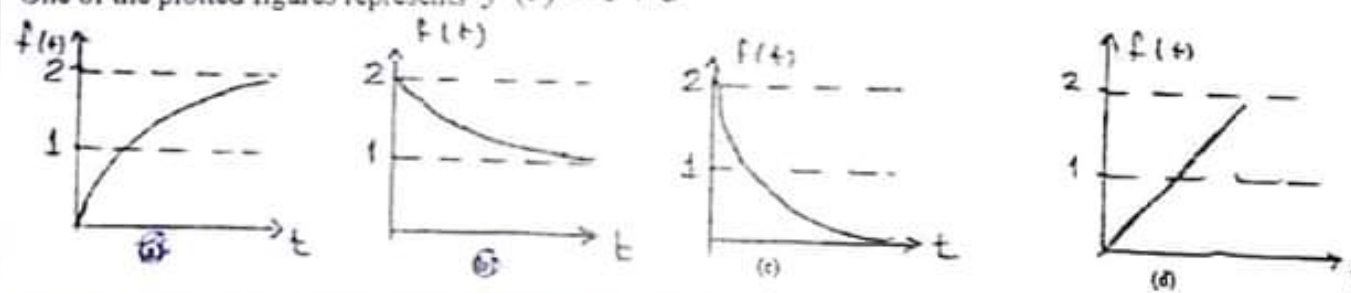
lecture time _____

Dr. Hussein Sarhan

Dr. Ibrahim Al-Abbas

Choose the most correct answer: Select the most correct answer from the alternatives given, and then write it in the table below.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
c	a	d	a	b	a	a	a	a	a	a	c	d	a	a	b	d	a	d	d

(1)	The output signal $C(s)$ equal the transfer function $G(s)$ [$C(s)=G(s)$], if the input to this system is: (a) unit step (b) unit ramp (c) unit impulse (d) ramp function
(2)	According to signal flow graph, which among the following represents the relationship between nodes by drawing a line between them? a. Branch b. Self-loop c. Semi-node d. Mesh
(3)	Given $Y(s) = \frac{(s+50)}{(s+10)^3}$. The residues are: (a) 40; 1, 0 (b) -40; 1, 0 (c) 20; -1, 0 (d) 50; 10, 0
(4)	One of the plotted figures represents $f(t) = 1 + e^{-2t}$ 
(5)	The transfer function of an LC series circuit, where $v_c(t)$ is the output and the applied voltage $v(t)$ is the input, is (a) $\frac{L}{LCs+1}$ (b) $\frac{1}{LCs^2+1}$ (c) $\frac{Cs}{LCs^2+1}$ (d) $\frac{1}{LCs+1}$

(6)	The poles of the output function $Y(s)$ are imaginary conjugate, the type of time function $y(t)$ is (a) exponential (b) damped ramp (c) undamped sinusoidal (d) damped sinusoidal
(7)	Inverse Laplace of $Y(s) = \frac{4}{(s+1)^2}$ is (a) $4e^{-t}$ (b) $4te^{-t}$ (c) te^{-t} (d) $4t^2$
(8)	The state variables of a system given by $y''(t) + 5y'(t) + 4y(t) = 3u(t)$ are (a) $x_1(t) = y(t); x_2(t) = y'(t); x_3(t) = y''(t)$ (c) $x_1(t) = 2y(t); x_2(t) = 4y'(t); x_3(t) = 5y''(t)$ (b) $x_1(t) = y(t); x_2(t) = r(t)$ (d) $x_1(t) = y(t); x_2(t) = y''(t)$
(9)	For the shown block diagram, the transfer function relating $C(s)$ and $R(s)$ is (a) $1 - G_1G_2 + G_2$ (b) $1 + G_1G_2 - G_2$ (c) $1 + G_1G_2 + G_2$ (d) $1 - G_1G_2 - G_2$
(10)	In the shown signal flow graph there are (a) three single loops (b) three single loops and two nontouching loops (c) four single loops and two nontouching loops (d) four single loops
11	In steam engine speed control system the feed back element is (a) power cylinder (b) gear box (c) pilot valve (d) control valve
12	One of the following is not a parameter of translation mechanical system (a) mass M (b) spring constant K (c) Applied force (d) friction factor
13	Transfer functions are commonly used when the control system is: (a) Nonlinear (b) Time varying (c) Multi input multi output (d) single input single output

4	A first-order dynamic system is represented by the following state equations $5x'(t) + x(t) = u(t)$, $y(t) = x(t)$ where $u(t)$ is the input, $y(t)$ is the output. The corresponding transfer function is <table> <tr> <td>(a) $G(s) = \frac{1}{1+5s}$</td><td>(b) $G(s) = \frac{5}{1+5s}$</td></tr> <tr> <td>(c) $G(s) = \frac{1}{s+5}$</td><td>(d) $G(s) = 5s+1$</td></tr> </table>	(a) $G(s) = \frac{1}{1+5s}$	(b) $G(s) = \frac{5}{1+5s}$	(c) $G(s) = \frac{1}{s+5}$	(d) $G(s) = 5s+1$
(a) $G(s) = \frac{1}{1+5s}$	(b) $G(s) = \frac{5}{1+5s}$				
(c) $G(s) = \frac{1}{s+5}$	(d) $G(s) = 5s+1$				
15	In level control system the Head (H) is equivalent to in electrical system (a) Current (b). Resistance voltage drop (c) Applied voltage (d) capacitor charge				
16	State space analysis is applicable even if the initial conditions are: a. Zero b. Non-zero c. Equal d. Not equal				
17	Which of the following transfer functions will have the greatest maximum overshoot? (a) $9/(s^2+2s+9)$ (b) $16/(s^2+2s+16)$ (c) $25/(s^2+2s+25)$ (d) $36/(s^2+2s+36)$				
18	Given a unity feedback system with $G(s) = K/s(s+4)$. What is the value of K for a damping ratio of <u>0.5</u>? (a) 1 (b) 16 (c) 4 (d) 2				
19	Given a unity feedback system with $G(s) = 9/\{s(s+4)\}$, the unit ramp time response of this system is (a) over damped (b) under damped (c) critically damped (d) none				
20	Open loop control can be used only if: (a) There are no external disturbances (b) There are no internal disturbances (c) If the relationship between the out put and the input is will known (d) All above				

Date:

Solutions

① C

② A

③ A

④ B

⑤ B

⑥ C

⑦ B

⑧ A

⑨ C

⑩ B

⑪ B

⑫ C

⑬ D

⑭ A

⑮ D

⑯ B

⑰ A

⑱ B

⑲ C

⑳ D

4) If the system has two real and equal poles, it is described as

(A) undamped (B) critically damped (C) underdamped (D) overdamped

5- A transfer function of first order system is given by:

$TF = \frac{10}{s+0.5}$, the time constant is: a) 0.5 b) 2 c) 10 d) None

4) If the system has two real and equal poles, it is described as

(A) undamped (B) critically damped (C) underdamped (D) overdamped

5- A transfer function of first order system is given by:

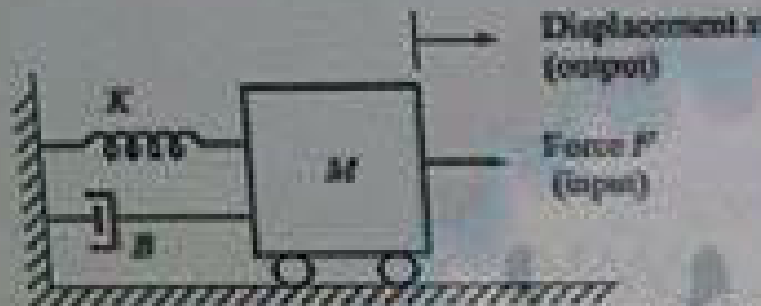
$TF = \frac{10}{s+0.5}$, the time constant is: a) 0.5 b) 2 c) 10 d) None

Q1) Select the most accurate answer

1) Choose the most accurate statement regarding the open and close loops.

- (A) Output of the open loop is dependent on reference signal
 (B) Close loop systems have less stability problems than open loops.
 (C) Open loop is easier to maintain
 (D) None

2- The system shown is of:



- (A) zero order (B) first order (C) second order (D) higher order

3) The partial fraction expansion of $Y(s) = \frac{s+2}{s^2(s+1)^2}$ is

- (A) $\frac{s+2}{s^2} + \frac{s+2}{s+1}$ (B) $\frac{s+2}{s^2} + \frac{s+2}{(s+1)^2}$ (C) $\frac{a_1}{s^2} + \frac{a_2}{s} + \frac{b_1}{(s+1)^2} + \frac{b_2}{s+1}$ (D) None

4) The inverse laplace transform of $Y(s) = \frac{1}{s^2(s+1)}$

- (A) $y(t) = t - 1 + e^{-t}$ (B) $y(t) = t^2 - t + e^{-t}$ (C) $y(t) = t^2 + t + e^{-t}$ (D) none

5) The positioning system of a printer can be modeled as

$$Y(s) = \frac{10(s+50)}{s^2 + 60s + 500} R(s).$$

where the input $R(s)$ represents the desired position and $Y(s)$ is the output position. If the input is a unit step input, the final value of the output is:

- (A) $y_{ss} = \lim_{t \rightarrow \infty} y(t) = 50$ (B) $y_{ss} = \lim_{t \rightarrow \infty} y(t) = 100$
 (C) $y_{ss} = \lim_{t \rightarrow \infty} y(t) = 1$ (D) none

6) Consider the differential equation

$$\ddot{y} + 2\dot{y} + y = u$$

Where $y(0) = \dot{y}(0) = 0$ and $u(t)$ is a unit step. The poles of this system are:
 A) $s_1 = -1, s_2 = -1$ B) $s_1 = -1, s_2 = -2$ C) $s_1 = 1j, s_2 = -1j$ D) None

Using the following matrices answer 7 and 8

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

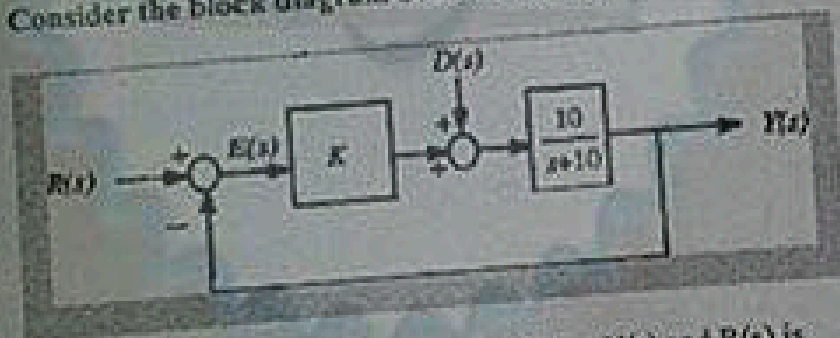
7) The state equations are

A) $\begin{cases} \dot{x}_1 = x_2 + u_2 \\ \dot{x}_2 = x_1 + u_1 \end{cases}$ B) $\begin{cases} \dot{x}_1 = x_2 - u_2 \\ \dot{x}_2 = x_1 + u_1 \end{cases}$ C) $\begin{cases} \dot{y}_1 = x_1 + u_1 \\ \dot{y}_2 = x_2 \end{cases}$ D) None

8) The output equations are

A) $\begin{cases} y_1 = x_1 + u_1 \\ y_2 = x_2 \end{cases}$ B) $\begin{cases} y_1 = x_2 + u_1 \\ y_2 = x_1 \end{cases}$ C) $\begin{cases} y_1 = x_1 + u_2 \\ y_2 = x_2 \end{cases}$ D) None

Consider the block diagram below for problems 9 and 10



9) The closed loop transfer function relating $Y(s)$ and $R(s)$ is

A) $\frac{10K}{s+10K+10}$ B) $\frac{10}{s+10K+10}$ C) $\frac{10K}{s+10}$ D) None

10) The closed loop transfer function relating $Y(s)$ and $D(s)$ is

A) $\frac{10K}{s+10K+10}$ B) $\frac{10}{s+10K+10}$ C) $\frac{10K}{s+10}$ D) None

Choose the correct answer using MATLAB.

Q1)

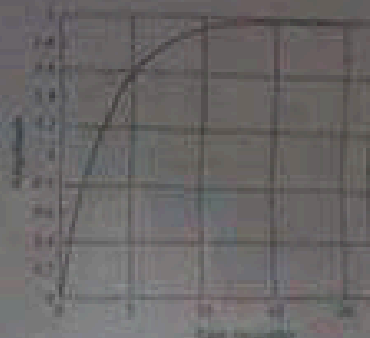
Closed-loop transfer function of a unity-feedback system is given by $T(s)/P(s) = 1/(2s + 1)$. Steady-state error to unit-ramp input is

- (A) ∞ (B) 1 (C) 1 (D) $1/2$

Q2)

The plot in Fig. shows the unit step response of a first-order system. Transfer function of the system is

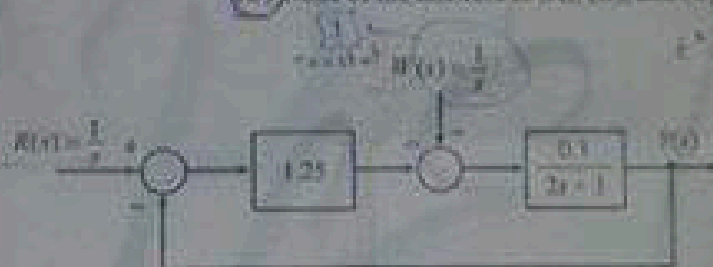
- (A) $\frac{1}{5s + 1}$ (B) $\frac{2}{15s + 1}$
(C) $\frac{2}{3s + 1}$ (D) $\frac{1}{15s + 1}$



Q3)

The system of Fig. has the steady-state error $e_{ss} = 1/2$

- (A) $\frac{1}{1.125}$ (B) $\frac{0.1}{1.125}$
(C) $\frac{1.1}{1.125}$ (D) None of the answers in (A), (B), and (C) is correct

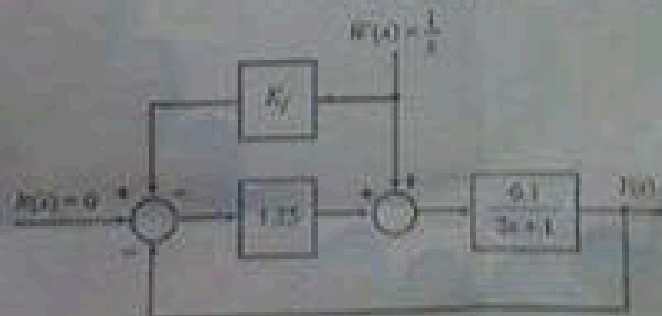


$$e_{ss} = \lim_{s \rightarrow 0} s E(s) = \lim_{s \rightarrow 0} s \left(\frac{1}{s} \right) \left(\frac{1}{1.125} \right) = \frac{1}{1.125}$$

Q4)

Consider the system of Fig. The steady-state error is zero for

- (A) $K_f = 0$ (B) $K_f = \frac{1}{1.25}$
(C) $K_f = 1.25$ (D) None of the answers in (A), (B), and (C) is correct



05.

$$G(x) = 20(x^2 + 5) \quad x \geq 0$$
(A) $k \geq 5$

494-5

(D) All the answers in (A), (B), and (C) are correct.

0204

$r(t) = K_1 + K_2 t + \frac{1}{6} t^3$. The steady-state error of the system is

eg: Input - output



Palladium

1811

100-2000

(D) None of the answers in (A), (B), and (C) is correct

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & c \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u$$

(A) $\frac{1}{(x+1)(x+2)}$

$$(B) \quad \frac{1}{s+1}$$
(C) $\frac{1}{x+3}$

(D) None of the answers in (A), (B), and (C) is correct

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	C	D	B	B	A	D

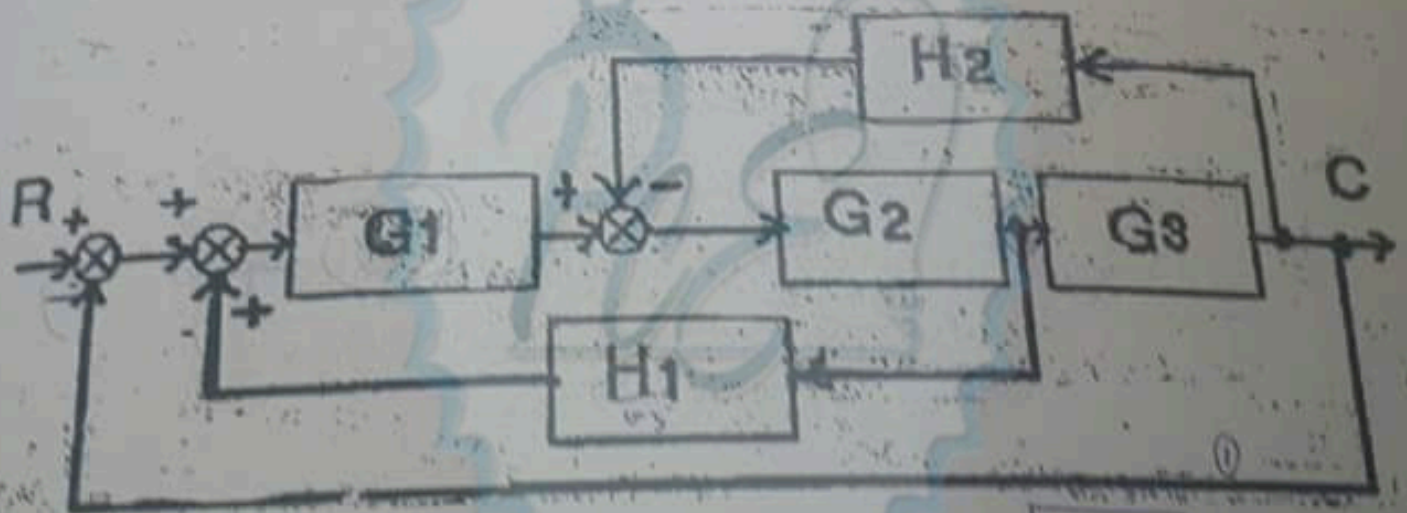
10/20

B C C B D B B

الحمد لله رب العالمين

Q2)(10 marks)

a- Reduce the following block diagram to find the transfer function of the system.





Second
Exam
20 Points

Instructors: Dr. Mahdi Al-Shamaseen
Dr. Mohammad Alkhdawaldah

Student Name: _____

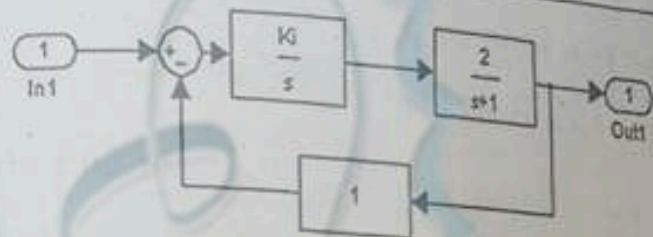
Select the most correct answer from the alternatives given, and then write it in the following table

1	2	3	4	5
B	A	C	B	C

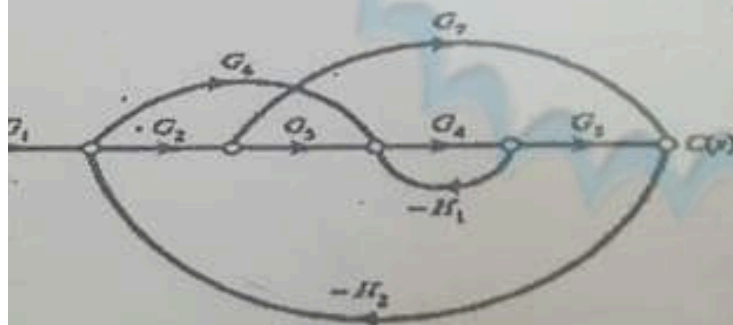
1. A unity feedback system has open-loop transfer function $G(s) = \frac{25}{s(s+6)}$, The time t_p at which the peak of the step-input response occurs is
- A- 11/7 sec B- 11/4 sec C- 11/14 sec. D- 11/28 sec.

2. If $K_i=2$, the damping ratio is

A] 0.5
B] 0.25
C] 0

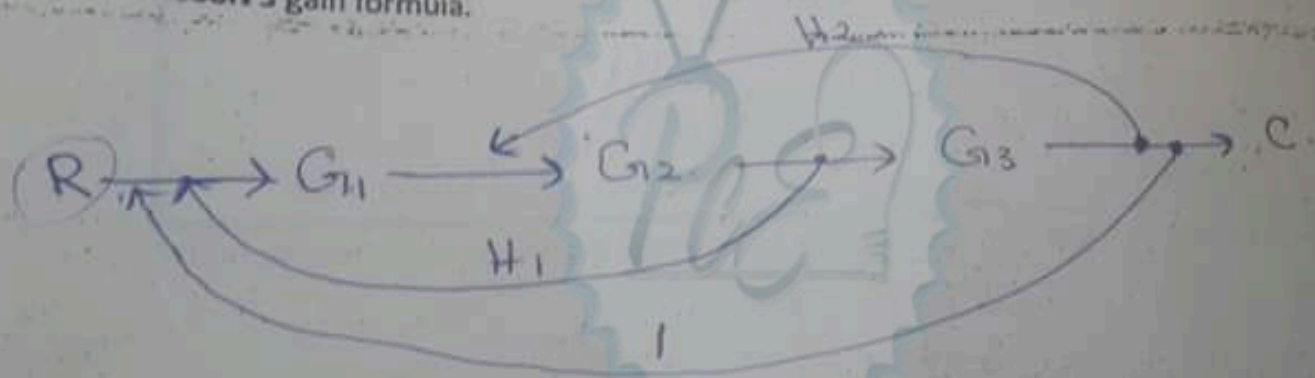


he given signal flow graph the number of closed loops is



- A] 2
B] 3
C] 4
D] 5

b- Convert the block diagram in (a) to signal flow graph and then, find the transfer function by Mason's gain formula.



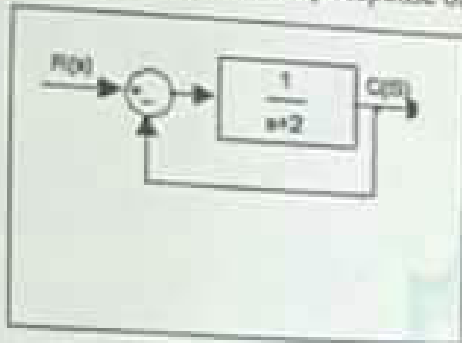
path = $G_1 G_2 G_3$

Q1 (16 marks) Select the most correct answer from the alternatives given, and then write it in the table below.

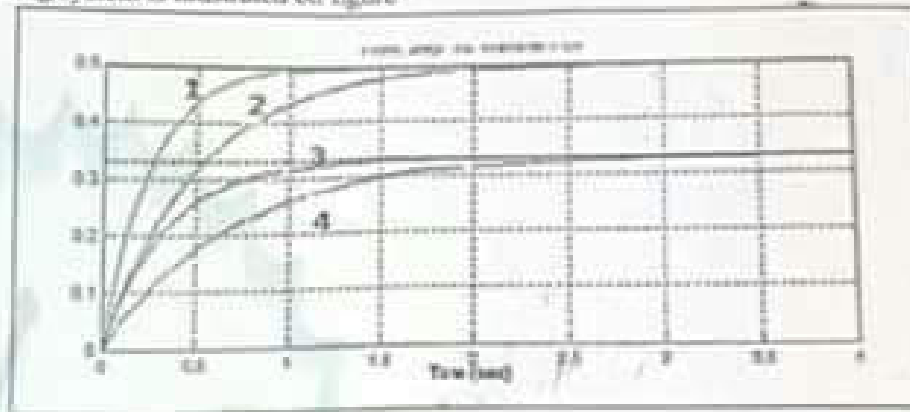
1- For most control systems the acceptable response should be
 A) over damped B) under damped C) Undamped D) critically damped

2- To eliminate the steady state error we use
 A) on-off controller B) P controller C) I controller D) D controller

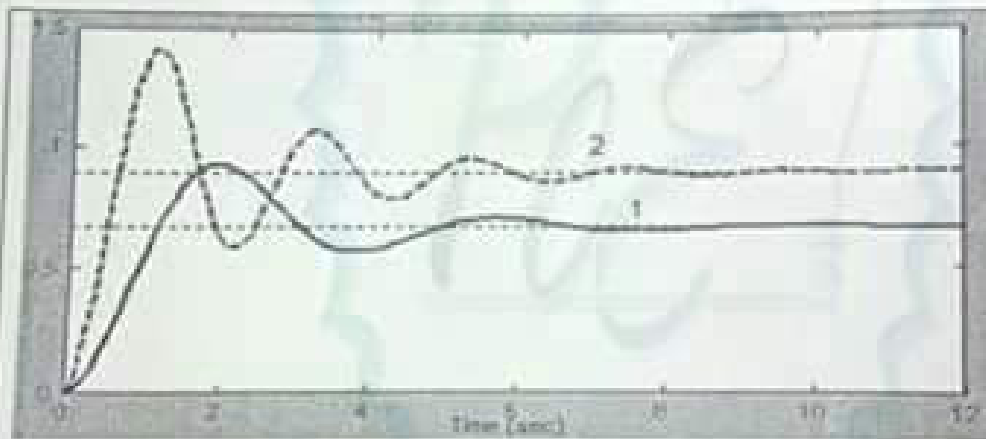
3- The unit step response of the following system is illustrated on figure



A) 4 B) 3 C) 2 D) 1

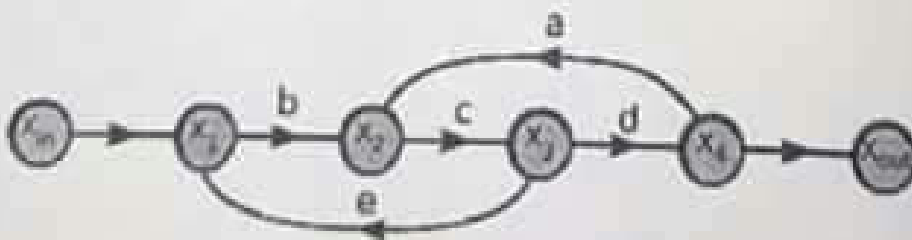


4- In the plotted figure: 1- is the unit step response of uncontrolled system, and 2- is the unit step response of controlled system. What type of controller is used?



A) P
 B) I
 C) PI
 D) PD

5- The determinant for the signal flow graph below is



- A) $1-bce-acd$ B) $1+bce+acd$ C) $1-bce-acd+bceacd$ D) $1+bce+acd-bceacd$

The characteristics equation of a closed loop control system is $s^3 + as^2 + bs + 2 = 0$. Answer 6 and 7

- 6- If $a=1, b=2$ the system is
A) stable B) critically stable C) unstable D) conditionally stable

- 7- The system is stable for
A) $a>0, b>0$ B) $a<0, b<0$ C) $ab>2$ D) $a>0, b>0, ab>2$

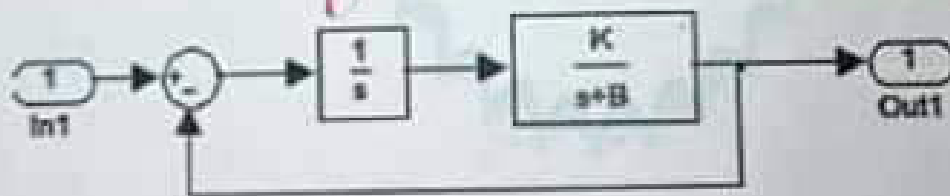
- 8- The mostly used controller in industry is
A) PID controller B) on-off controller C) PI controller D) PD controller

1	2	3	4	5	6	7	8
D	C	B	D	A	A	D	C

Q2) (4 marks) For the system shown below $K=4$ and $B=2$. Determine the damping ratio and the damped frequency. Write your answer in the given space.

Damping ratio =

Damped frequency =



- 1) In steam engine speed control system the feed back element is
 A) pilot valve B) gear box C) power cylinder D) speed of rotation

2) The residues in partial fraction expansion of $F(s) = \frac{4s^2 + 8s + 4}{s(s+2)^3}$ is

- A) 1, 2, 3 B) -1, 2, 3 C) 1, -2, 3 D) 1, 2, -3

3) The inverse Laplace transform of $F(s) = \frac{s+3}{s(s^2 + 5s + 10)}$ is

- A) $1 - e^{-5} \cos t$ B) $1 + e^{-5} \cos t$ C) $1 - e^{-5} \cos t$ D) $e^{-5} \cos t$

4) in control systems feed back is an operation used to

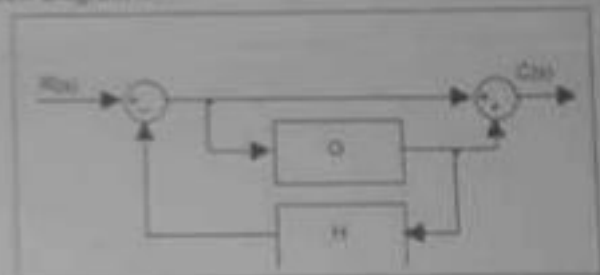
- A) fast the response B) increase the stability C) improve system dynamics
 D) reduce the difference between the output of a system and some reference input

5) Transfer function approach usually used when the system is

- A) single input multi output B) single input single output
 C) multi input single output D) multi input multi output

6) the transfer function from the shown block diagram is

- A) $\frac{1+G}{1+GH}$ B) $\frac{1+G}{1-GH}$
 C) $\frac{G}{1+GH}$ D) none

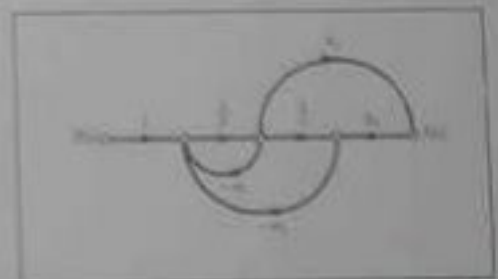


7) In RLC series circuit the applied voltage is the input and the voltage drop across the resistance is the output, the transfer function of this system is

- A) $\frac{Cs}{LCs^2 + RCs + 1}$ B) $\frac{Ls}{LCs^2 + RCs + 1}$
 C) $\frac{Rs}{LCs^2 + RCs + 1}$ D) $\frac{RCs}{LCs^2 + RCs + 1}$

8) the transfer function of the given signal flow graph is

- A) $\frac{b_1s + b_2}{s^2 + a_1s + a_2}$ B) $\frac{b_1s + b_2}{s^2 + a_1s + a_2}$



$$C) \frac{b_1 s + b_0}{s^2 + a_1 s + a_0} \quad D) \frac{b_1 s - b_0}{s^2 + a_1 s + a_0}$$

A control system is given as

$$\dot{x}_1'(t) = x_1(t) + x_2(t) + u_1(t) + u_2(t)$$

$$\dot{x}_2'(t) = x_1(t) + u_1(t)$$

$$y_1(t) = x_1(t) + u_2(t)$$

$$y_2(t) = u_1(t) + u_2(t)$$

Answer 9 and 10

9) the state matrix **A** is

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

10) the output matrix **D** is

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$



D -

$$2) Y(s) = \frac{4s^2 + 8s + 4}{s(s+2)^2}$$

$$\frac{A}{s} + \frac{B}{(s+2)} + \frac{C}{(s+2)^2} =$$

$$A(s+2)(s+2)^2 + B(s)(s+2)^2 + Cs = 4s^2 + 8s + 4$$

$$A(s^2 + 4s + 4) + Bs^2 + 2Bs + Cs = 4s^2 + 8s + 4$$

$$As^2 + Bs^2 + 2Bs + Cs = 4s^2 + 8s + 4$$

$$As^2 + Bs^2 = 4s^2 \rightarrow A + B = 4 \rightarrow B = 3$$

$$A4s + 2Bs + Cs = 8s \rightarrow 4A + 2B + C = 8$$

$$4A = 4 \rightarrow A = 1 \quad C = 8 - 10 = -2 \quad \#C$$

$$3) Y(s) = \frac{s+3}{s(s^2+6s+10)} \rightarrow \frac{A}{s} + \frac{Bs+C}{s^2+6s+10}$$

$$A(s^2+6s+10) + B(s+C)s = s+3$$

$$A = 0.3 \rightarrow A+B=0$$

$$6A+C=1$$

$$10A=3 \Rightarrow B = -0.3, C = -0.8$$

$$\frac{0.3}{s} + \frac{-(0.3s+0.8)}{s^2+6s+10+9-9}$$

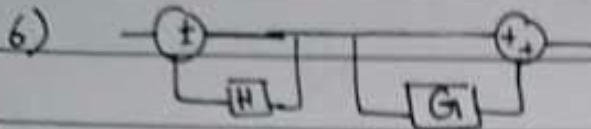
$$\frac{0.3}{s} + \frac{-(0.3s+0.8) + 0.1 - 0.1}{(s+3)^2 + 1}$$

$$0.3 - 0.3(\cos t \cdot e^{-3t} - 0.1 \sin t \cdot e^{-3t})$$

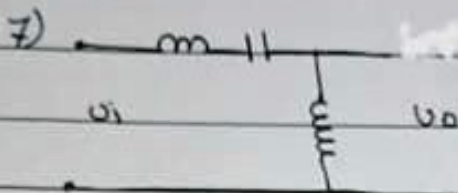


4) D

5) B



$$\frac{1}{1+HG} \cdot (1+G) \quad \#A$$



$$u_1 = L \frac{di}{dt} + \frac{1}{C} \int i dt + R i(A)$$

$$u_2(s) = L(s) + \frac{1}{Cs} I(s) + R I(s)$$

$$\Rightarrow \frac{I(s)}{u_2(s)} = \frac{1}{L(s) + \frac{1}{Cs} + R} \Rightarrow \frac{Cs}{CsLs + 1 + CSR}$$

$$\frac{u_0}{s} = R I s \rightarrow \frac{u_0}{I s} = R$$

$$\frac{u_0}{u_2} = \frac{RCS}{CsLs + 1 + CSR} \quad \#D$$

8) $P_1 =$

$$1 \quad \frac{1}{s} \quad \frac{1}{s} \quad b_2$$

$P_2 =$

$$1 \quad \frac{1}{s} \quad b_1$$

9) #B

10) #D

$$L_1 = -a_1 \frac{1}{s}, \quad L_2 = -a_2 \frac{1}{s} \frac{1}{s}$$

$$\Delta = 1 - (-a_1 \frac{1}{s} + -a_2 \frac{1}{s} \frac{1}{s}) + 0$$

$$\Delta = 1 + a_1 \frac{1}{s} + a_2 \frac{1}{s} \frac{1}{s}$$

$$\Delta_1 = 1$$

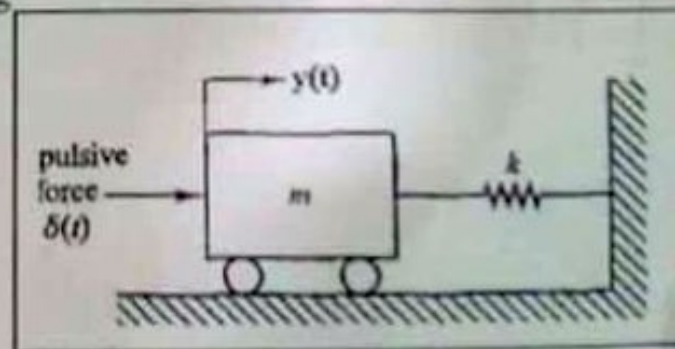
$$\Delta_2 = 1$$

$$t.f = \frac{P_1 \Delta_1 + \Delta_2 P_2}{\Delta} = \frac{1 \cdot 1 + 1 \cdot b_1}{1 + a_1 \frac{1}{s} + a_2 \frac{1}{s}}$$

$$= \frac{(\frac{1}{s} b_2 + b_1 \frac{1}{s})}{(1 + a_1 \frac{1}{s} + a_2 \frac{1}{s})} s^e = \frac{b_2 + b_1 s}{s^2 + a_1 s + a_2} \quad \#A$$

1	2	3	4	5	6	7	8	9	10	11	12	13	14
B	A	D	D	C	A	B	A	A	B	A	C	D	D

1. Let $F(s) = \frac{1}{s(s+1)}$, The final value of $f(t)$ is: $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s) = \lim_{s \rightarrow 0} \frac{1}{s+1} = 1$
 a) 0 ☒ b) ∞ c) 1 d) None
2. $y(t) = e^{-t} - e^{-2t}$, Then $Y(s)$ will be
☒ A. $\frac{1}{s^2+1}$ B. $\frac{-2}{s^2-1}$ C. $\frac{+2}{s^2-1}$ D. None
3. Open-loop control can be used, in practice, only if:
 A. the relationship between the input and output is known B. there are no internal disturbances.
 C. there are no external disturbances ☒ D. all above
4. The transfer function of an L C series circuit, where $v_c(t)$ is the output and the applied voltage $v(t)$ is the input, is
 A. $\frac{L}{LCs+1}$ ☒ B. $\frac{1}{LCs^2+1}$ C. $\frac{Cs}{LCs^2+1}$ ☒ D. $\frac{1}{LCs+1}$
5. The number of state equations in spring-mass-dashpot linear mechanical system is
 A. 4 B. 3 ☒ C. 2 D. 1
6. The output of the main comparator in a feedback control system is called
☒ A. error signal B. Feedback signal C. output signal. D. Difference signal
7. The major disadvantage of a feedback system may be
 A. Inaccuracy ☒ B. instability C. Unreliability D. inefficiency
8. For the system shown, the mathematical model relating The output ($y(t)$) and the input ($\delta(t)$) is
☒ A. $my''(t) + fy'(t) + Ky(t) = \delta(t)$
 B. $my''(t) + Ky(t) = \delta(t)$
 C. $my''(t) + fy'(t) + Ky(t) = 0$
 D. $my''(t) + Ky(t) = 0$
9. The state variables of a system given by $y'''(t) + 5y''(t) + 4y'(t) + 2y(t) = 3u(t)$ are
☒ A. $x_1(t) = y(t); x_2(t) = y'(t); x_3(t) = y''(t)$ B. $x_1(t) = 2y(t); x_2(t) = 4y'(t); x_3(t) = 5y''(t)$
 C. $x_1(t) = y(t); x_2(t) = r(t)$ D. $x_1(t) = y(t); x_2(t) = y''(t)$
10. The steady state error in the unit ramp response of first order standard system is
 A. 0 ☒ B. T C. 2T D. 3T



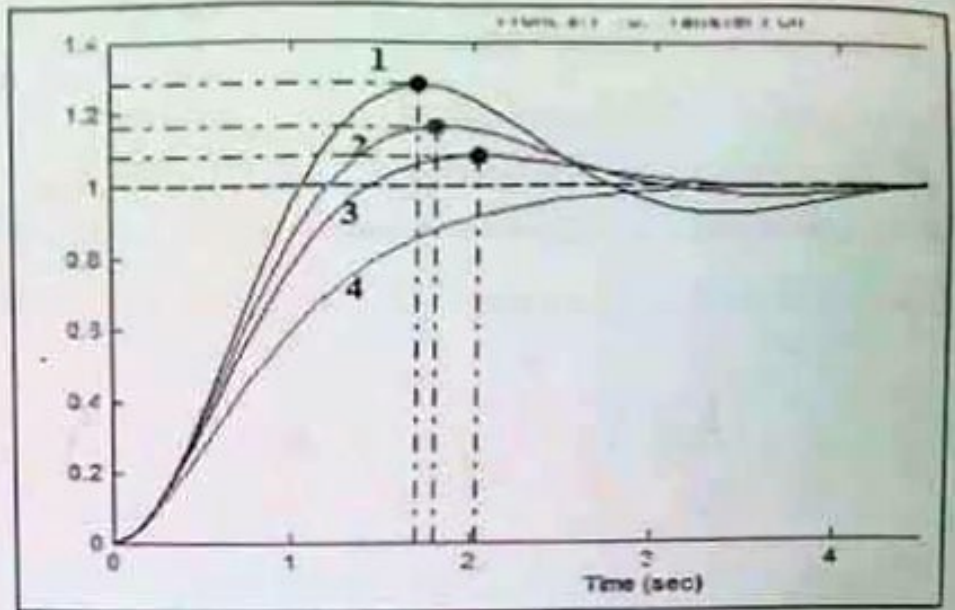
Depend on the following figure to solve 11 & 12.

11. The time response of a second Order system given by

$$\frac{C(s)}{R(s)} = \frac{4}{s^2 + 2s + 4}$$

is denoted as

- ☒ A. 1 B. 2 ☐ C. 3 D. 4



12. The unacceptable response with respect to the rise time is denoted by

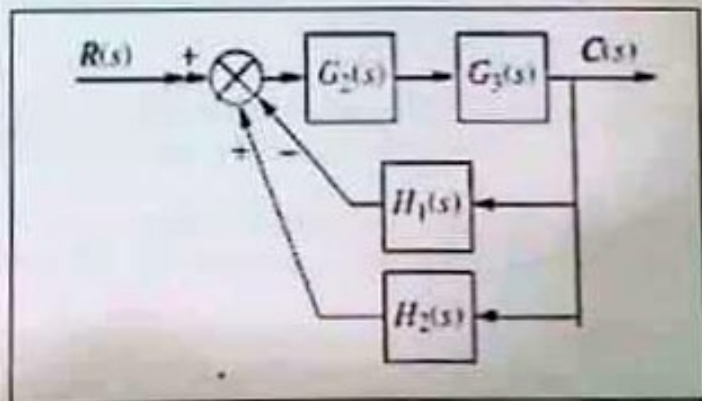
- ☒ A. 1 B. 2 C. 3 D. 4

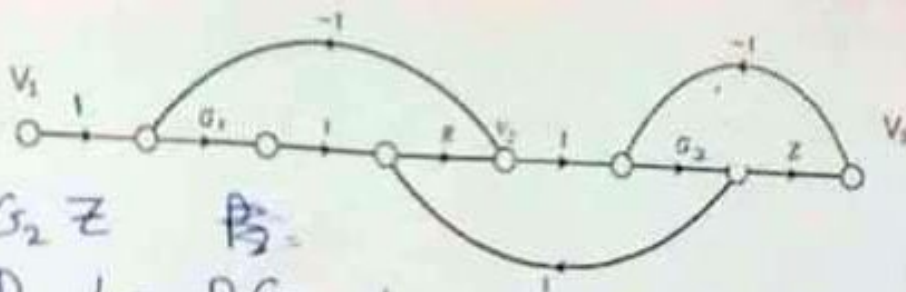
13. The partial fraction of $y(s) = \frac{1}{s^2(s+1)^2}$ is

- A. $y(s) = \frac{1}{s^2(s+1)^2}$ B. 2 $y(s) = \frac{a}{(s+1)^2} + \frac{b}{s^2}$
☒ C. $y(s) = \frac{\alpha_1 s + \alpha_2}{(s+1)^2} + \frac{b}{s^2}$ ☐ D. $y(s) = \frac{\alpha_1 s + \alpha_2}{(s+1)^2} + \frac{\beta_1 s + \beta_2}{s^2}$ ← ans D

14. The open loop transfer function from the given block diagram is

- A. $G_2 G_3 H_1 H_2(s)$
 B. $G_2 G_3 H_1(s)$
 C. $G_2 G_3 H_2(s)$
☒ D. $G_2 G_3 (H_2(s) - H_1(s))$





$$P_1 = G_1 R G_2 Z$$

$$L_1 = -G_1 R \quad L_2 = -R G_2 \quad L_3 = -G_2 Z$$

untouch loop

$$L_1 L_2 = -G_1 R^2 \quad L_1 L_3 = +G_1 G_2 R Z \quad L_2 L_3 = -R G_2^2 Z$$

$$\Delta = 1 - [-G_1 R + R G_2 - G_2 Z] + [-G_1 R^2 + G_1 G_2 R Z - R G_2^2 Z]$$

$$\Delta_1 = 1$$

$$\frac{V_3}{V_1} = \frac{\Delta_1 \times P_1}{\Delta} = \frac{1 \times G_1 R G_2 Z}{1 - [-G_1 R + R G_2 - G_2 Z] + [-G_1 R^2 + G_1 G_2 R Z - R G_2^2 Z]}$$

Q3) (6 marks) Assume zero initial conditions, solve the following differential equation using Laplace transform

$$y''(t) + 3y'(t) + 2y(t) = 1$$

$$s^2 y(s) + 3s y(s) + 2y(s) = \frac{1}{s}$$

$$y(s) [s^2 + 3s + 2] = \frac{1}{s}$$

$$y(s) = \frac{1}{s[s^2 + 3s + 2]} \Rightarrow \frac{1}{s(s+2)(s+1)} \Rightarrow \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+1}$$

$$A(s+2)(s+1) + B(s+1)s + C(s+2)s = 1$$

$$A = \frac{1}{2} \quad B = \frac{1}{2} \quad C = -1$$

$$\Rightarrow \frac{1}{2} + \frac{1}{2} e^{-2t} - e^{-t}$$



$$1) F(s) = \frac{1}{s(s+1)} \Rightarrow \lim_{s \rightarrow 0} F(s) = \lim_{s \rightarrow 0} s \frac{1}{s(s+1)} = 1 \quad \# \textcircled{C}$$

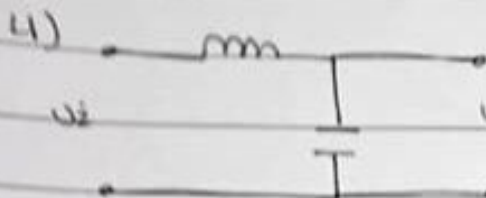
$$\text{OR } F(s) = \frac{A}{s} + \frac{B}{s+1} \rightarrow A(s+1) + B(s) = 1, A=1, B=-1$$

$$F(s) = \frac{1}{s} + \frac{-1}{s+1} \quad F(t) = 1 - e^{-t}$$

$$\lim_{s \rightarrow \infty} F(s) = 1 \quad \# \textcircled{C}$$

$$2) Y(t) = \frac{-t}{e} e^t \sim Y(t) = -2 \sinh t \rightarrow Y(s) = \frac{-2}{s^2+1} \quad \# \textcircled{B}$$

$$3) \# \textcircled{D}$$



$$U_i = L \frac{di}{dt} + \frac{1}{C} \int I(t) dt$$

$$U_i(s) = L(s) I(s) + \frac{1}{Cs} I(s)$$

$$\frac{U_i(s)}{U_i(s)} = \frac{1}{2s + \frac{1}{Cs}} \rightarrow \frac{Cs}{Cs^2 + 1}$$

$$\Rightarrow U_o = \frac{1}{C} \int I(t) dt \Rightarrow U_o(s) = \frac{1}{Cs} I(s)$$

$$\frac{U_o(s)}{U_i(s)} = \frac{1}{Cs^2 + 1} = \frac{1}{Cs^2 + 1} \quad \# \textcircled{B}$$

$$5) \# \textcircled{C}$$

$$6) \# \textcircled{A}$$

$$7) \# \textcircled{B}$$

$$8) m \ddot{y}(t) + k \dot{y}(t) = \delta(t) \quad \# \textcircled{B}$$

$$F = k \dot{y}(t)$$

الذنب

$$a) \ddot{y}(t) + 5 \dot{y}(t) + 4 y(t) + 2 y(t) = 3 u(t)$$

$$x_1 = y(t) = x_2$$

$$x_2' = \dot{y}(t) = x_3$$

$$x_3' = \ddot{y}(t) = -5 \dot{y}(t) - 4 y(t) - 2 y(t) + 3 u(t) \quad \# \textcircled{A}$$



Power and Communication Engineering

10) $r(t) = t$ ✗ (B)

11) ✗ B $\rho_{\text{out}} = \frac{1}{\omega} = 1.8 \approx 2$

~~Ready~~ $t_p = \frac{\pi}{\omega} = 1.8 \approx 2$

12) ✗ (D) H_{zero}

13) $\frac{1}{s^2(s+1)^2} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1} + \frac{D}{s+1}$ ✗ (D)

14) ✗ D Feedback $\rightarrow$ open loop

Q2) $P_1 = 1 \quad G_1 \quad R \quad G_2 \quad Z$

$L_1 = G_1 R - 1$

$L_2 = R G_2 (1)$

$L_3 = G_2 Z - 1$

non tech loop

$L_4 L_3 = -G_1 R - G_2 Z$

$\Delta = 1 - \sum \text{loop} + \sum \text{non tech}$

$= 1 - (-G_1 R - G_2 Z + R G_2) + G_1 G_2 R Z$

$= 1 + G_1 R + G_2 Z - R G_2 + G_1 G_2 R Z$

$\Delta_1 = 1$

$\frac{V_o}{V_i} = \frac{\Delta R_1}{\Delta} = \frac{1 G_1 G_2 R Z}{1 + G_1 R + G_2 Z - R G_2 + G_1 G_2 R Z}$

Q3) $\ddot{y}(t) + 3\dot{y}(t) + 2y(t) = 1$

Sol: $s^2 y(s) + 3s y(s) + 2y(s) = \frac{1}{s}$

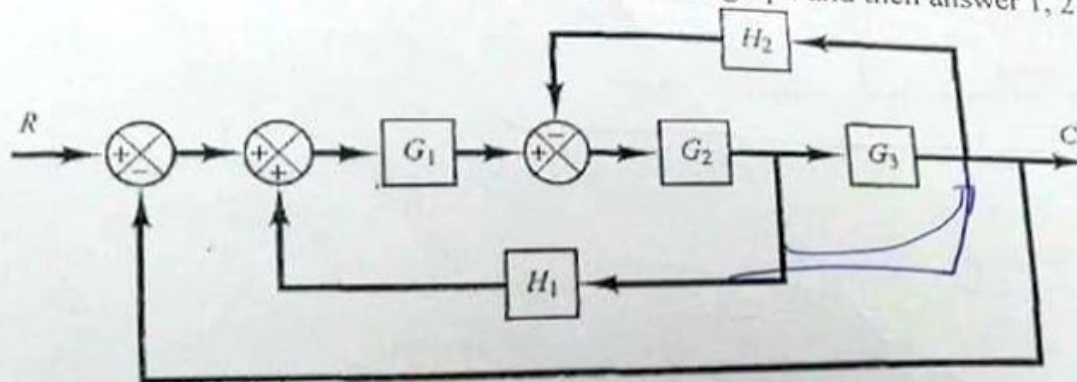
$y(s) = \frac{1}{s} \cdot \frac{1}{s^2 + 3s + 2} = \frac{1}{s} \cdot \frac{1}{(s+2)(s+1)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+1}$

$A(s+2)(s+1) + B s(s+1) + C (s+2)(s) = 1 \quad | A = \frac{1}{2}, B = \frac{1}{2}, C = -1$

$y(s) = \frac{\frac{1}{2}}{s} + \frac{\frac{1}{2}}{s+2} + \frac{-1}{s+1}$

$y(t) = \frac{1}{2} + \frac{1}{2} e^{-2t} - e^{-t}$

Convert the block diagram below to a signal flow graph and then answer 1, 2 & 3. (35 marks)



1- The number of non-touching loops (two at a time) is:
a) 2 b) 1 **c) 0** d) None

2- The determinant is equal:
a) $1 - G_1 G_2 H_1 - G_2 G_3 H_2 + G_1 G_2 G_3$ **b) $1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3$** c) 1 d) None

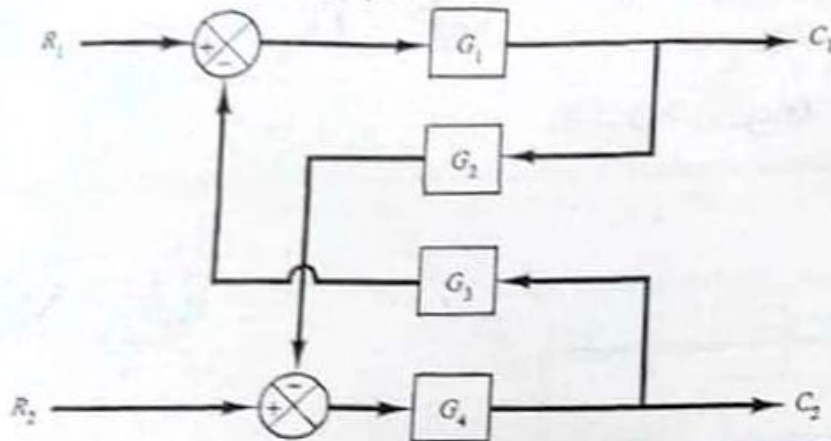
3- The transfer function $\frac{C(s)}{R(s)}$ is:

a) $\frac{G_1 G_2 G_3}{1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3}$ b) $\frac{G_1 G_2 H_2}{1 - G_1 G_2 H_1 + G_2 G_3 H_2 + G_1 G_2 G_3}$ c) $\frac{G_1 G_2 G_3}{1 - G_1 G_2 H_1 - G_2 G_3 H_2 + G_1 G_2 G_3}$ d) None

4- The determinant of the forward path $G_1 G_2 G_3$ is:
a) 1 b) $1 - H_1 H_2$ c) $1 + H_1 H_2$ d) None

5- The figure shown below represents a linear time invariant system, the transfer function C_1/R_2 of this system is

- a) $-\frac{G_1 G_3 G_4}{1 - G_1 G_2 G_3 G_4}$ b) $-\frac{G_1 G_3 G_4}{1 + G_1 G_2 G_3 G_4}$ c) $\frac{G_1 G_3 G_4}{1 + G_1 G_2 G_3 G_4}$ d) None



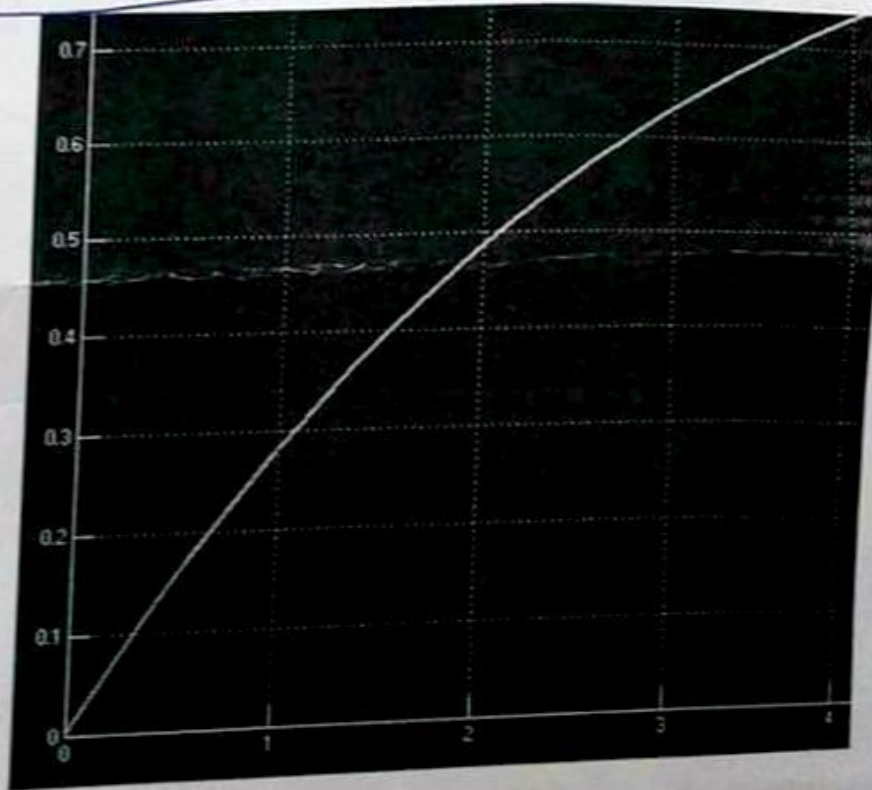
The adjacent figure shows the unit step time response of first order system. Use this figure to answer 6 & 7.

6- The time constant is

- a) 1 sec b) 2 sec. c) 3 sec. d) 4 sec

7- The settling time is:

- a) 6 sec b) 8 sec c) 17 sec d) 12 sec



8- One of the following is not true for Open loop control system

- a) Less expensive than close loop systems. b) less stable than close loop systems c) Simpler to construct than close loop. d) None

9- The function $F(s) = \frac{s+4}{s^2(s+2)^2(s+5)}$ can be expanded with partial fraction as follows:

a) $\frac{a}{s} + \frac{a}{s^2} + \frac{b}{s+2} + \frac{b}{(s+2)^2} + \frac{c}{s+5}$ b) $\frac{a}{s} + \frac{b}{s^2} + \frac{c}{s+2} + \frac{d}{(s+2)^2} + \frac{e}{s+5}$

~~c) $\frac{as+b}{s^2} + \frac{as+b}{(s+2)^2} + \frac{as+b}{(s+5)}$~~ d) None

10- The inverse Laplace transforms of $F(s) = \frac{2s+3}{s^2+4s+13}$ is: A) $F(s) = e^{-2t}(2\cos 3t - \sin 3t)$
B) $f(t) = e^{-2t}(2\cos 3t - \frac{1}{3}\sin 3t)$ C) $f(t) = e^{-2t}(2\cos 3t + \frac{1}{3}\sin 3t)$ D) None.

11- The initial value of $F(s) = \frac{2}{s(s+1)}$ is:

- a) 1 b) 2 c) 0 d) 4

For an RC series circuit, the input voltage is $e_i(t)$ and the output is the voltage across the capacitor. Use this information to answer 12, 13 & 14.

12- The Transfer function is:

a) $\frac{1}{RCS+1}$ b) $\frac{CS}{RCS+1}$ c) $\frac{R}{RCS+1}$ d) None

$$RC \frac{di}{dt} + \frac{1}{C} \int i dt = e_i$$

$$e_o = \frac{1}{C} \int i dt$$

13- The differential equation that describes the system is:

a) $e'_o(t) = \frac{e_i(t) + e_o(t)}{RC}$ b) $e'_o(t) = \frac{e_i(t) + e'_o(t)}{RC}$ c) $e'_o(t) = \frac{e_i(t) - e_o(t)}{RC}$ d) None

14- For state space representation, the matrix A =

a) $[\frac{1}{RC}]$ b) $[-\frac{1}{RC}]$ c) $[-1]$ d) None

1	2	3	4	5	6	7	8	9	10	11	12	13	14
c	b	a	d	c	c	a	d	d	A	C	D	C	b

22.5

Q2) (5 marks) Solve the following differential equation: $X''(t) + 2X'(t) + 10X(t) = e^{-t}$
 $X'(0) = 0, X(0) = 0$



1) 0 $\neq$ C

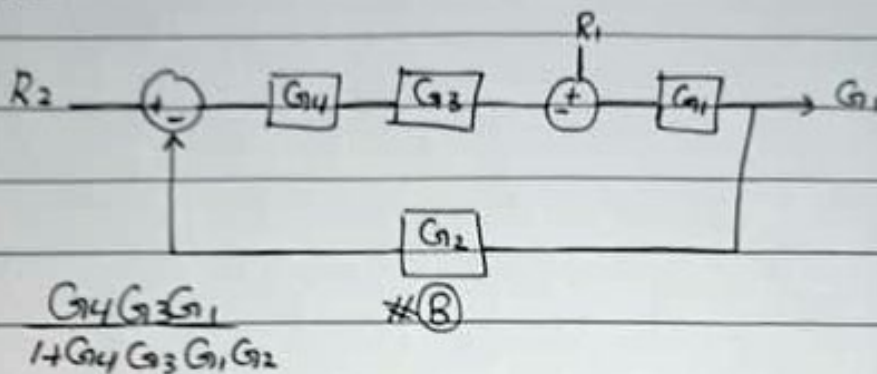
2) $\Delta = 1 - \sum \text{Loops} + \sum \text{non touching}$

$\Delta = 1 - G_1 G_2 H_2 + G_2 G_1 H_1 + G_1 G_2 G_3$ $\neq$ b

3) $\frac{G_1 G_2 G_3}{1 - G_1 G_2 H_1 + G_2 G_1 H_2 + G_1 G_2 G_3}$ $\neq$ a

4) $\Delta_{G_1 G_2 G_3} = 1$ $\neq$ a

5) $\frac{G_1}{R_2}$



6) $\gamma = 0.63$ steady state = 0.68

$\tau = 3 \text{ sec}$ $\neq$ C

7) 12 sec $\neq$ D $\rightarrow T_s = 4$ $T_{\text{constant}} = 12 \text{ sec}$

8) $\neq$ B

9) $F(s) = \frac{s+4}{s^2(s+2)^2(s+5)}$

$= \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+2} + \frac{D}{(s+2)^2} + \frac{E}{s+5}$ $\neq$ B

10) $F(s) = \frac{2s+3}{s^2+4s+13}$

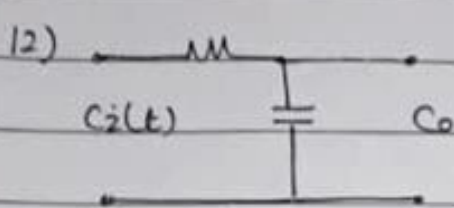
Sol: $\frac{2s+3}{s^2+4s+13} = \frac{2s+3}{s^2+4s+13.4-4} = \frac{2s+3}{s^2+4s+9+4}$



$$= \frac{2s+3}{(s+2)^2+9} = \frac{2s+3+1-1}{(s+2)^2+9} = \frac{2(s+2)-1}{(s+2)^2+9}$$

$$= \frac{2(s+2)}{(s+2)^2+9} - \frac{1}{(s+2)^2+9} = 2 \cos 3t \cdot e^{-2t} - \frac{1}{3} \sin 3t \cdot e^{-2t} \quad \# B$$

11) *C



$$c_i(t) = R I(t) + \frac{1}{C} \int I(t) dt$$

$$C_i(s) = R I(s) + \frac{1}{Cs} I(s)$$

$$C_i(s) = I(s) \left(R + \frac{1}{Cs} \right) \rightarrow \frac{I(s)}{C_i(s)} = \frac{1}{CsR+1}$$

$$\frac{I(s)}{C_i(s)} = \frac{C(s)}{CsR+1}$$

$$C_o(t) = \frac{1}{C} \int I(t) dt \rightarrow C_o(s) = \frac{1}{Cs} I(s)$$

$$\frac{C_o(s)}{I_o(s)} = \frac{1}{Cs} \Rightarrow \frac{C_o(s)}{C_i(s)} = \frac{1}{CsR+1} \quad \# A$$

13) $C_o = \frac{1}{C} \int I(t) dt \Rightarrow C_o' = \frac{1}{C} I(t)$

$$I(t) = \frac{C_o' - C_o}{R} \Rightarrow C_o = \frac{C_o' - C_o}{R} \quad \# C$$

14) $C_i(t) = R I(t) + \frac{1}{C} \int I(t) dt$

$$x_1 = \int I(t) dt$$

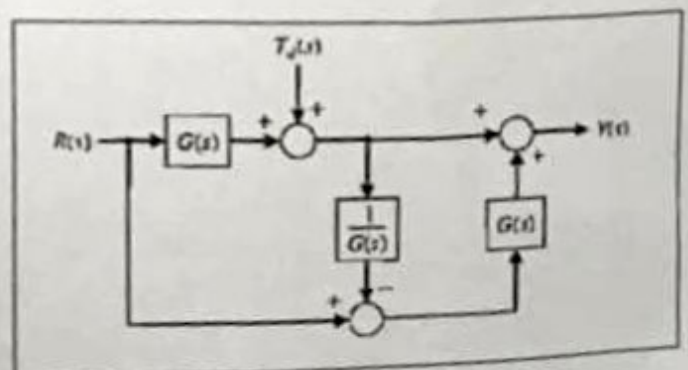
$$x_1' = I(t) = \frac{C_i(t)}{R} - \frac{1}{CR} \int I(t) dt$$

$$[x_1'] = \overset{A}{\begin{bmatrix} -1 \\ CR \end{bmatrix}} [x_1] + \overset{B}{\begin{bmatrix} 1 \\ R \end{bmatrix}} [C_i]$$

$$\# A \begin{bmatrix} -1 \\ CR \end{bmatrix} \quad \# B$$

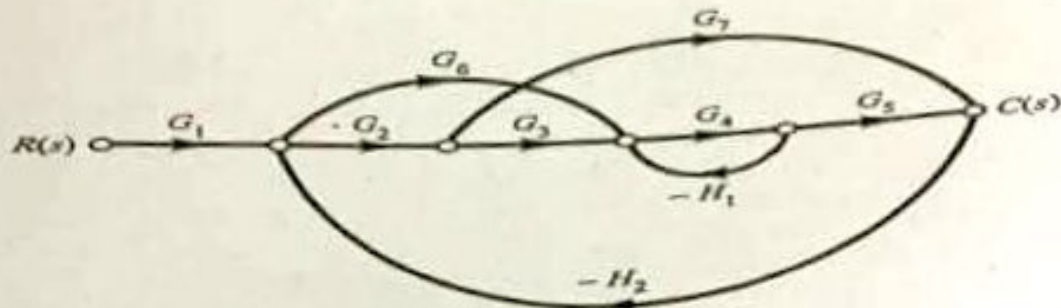
1. The system given by $m(t)y''(t) + ky(t) = u(t)$ is
☒ A. Nonlinear B. time invariant C. Time varying ☒ D. linear
2. The transfer function of an L C series circuit, where $v_c(t)$ is the output and the applied voltage $v(t)$ is the input, is
 A. $\frac{1}{LCs^2 + 1}$ ☒ B. $\frac{Cs}{LCs^2 + 1}$ C. $\frac{1}{LCs + 1}$ D. $\frac{L}{LCs + 1}$
3. The state space model of an translation mechanical system, where displacement is the output and the applied force is the input, includes an input matrix $B(s)$ with
☒ A. 1×1 B. 2×1 C. 1×2 D. 2×2
4. The inverse Laplace transforms of $Y(s) = \frac{10(s+4)}{s^2 + 8s + 15}$ is
 A. $y(t) = 10e^{-5t} + 10e^{-3t}$ B. $y(t) = 10e^{3t} + 10e^{-3t}$
 C. $y(t) = 10e^{3t} - 10e^{-3t}$ ☒ D. none
5. one of the following is an advantage of closed loop system
 A. low coast B. easy to maintain C. always stable ☒ D. high accuracy
6. "feed back" is operation used to:
 A. measure the output signal B. generate feed back signal
☒ C. reduce the difference between the output and reference input
 D. determine the feed back block transfer function
7. In speed control system the feedback element is
☒ A. pilot valve B. control valve
☒ C. gear box D. power cylinder

8. The transfer function relating $Y(s)/R(s)$ is



A] $-G(s)$	B] $1+G(s)$
C] $G(s)$	D] $G(s) + T_d(s)$

9. The number of nontouching loops in the signal flow graph is



A] 0	B] 2
C] 3	D] 4

A first-order system is represented by the differential equation $y'(t) + 4y(t) = 2u(t)$

Answer 10 and 11

10. the time constant of this system is

A. 2 B. 4 C. 0 D. 0.25

11- the steady state value of the output is

A. 0 B. 1 C. 2 D. 0.5

For a second order system given as $\frac{C(s)}{R(s)} = \frac{1}{s^2 + 5s + 4}$

answer 12 and 13

12 the unit step response of this system is

A. over damped B. under damped C. undamped D. critically damped

13. the damping ratio is

A. 0.8 B. 1 C. 0.25 D. 2



(Institution X)

1) One of the following is an advantage of closed loop.

- A) Low cost B) easy to maintain C) always stable
D) high accuracy.

2) The system given by $m(t) \ddot{y}(t) + ky(t) = u(t)$ is

- A) Non linear **B) Time varying** C) Time invariant D) Linear

3) The Transfer Function of Ls series circuit where $I(t)$ is the output and $V(t)$ is the input.

- A) $\frac{1}{Ls^2+1}$ **B) $\frac{Cs}{Ls^2+1}$** C) $\frac{1}{Ls+1}$ D) $\frac{1}{Ls+1}$

4) The state space model of an transfer mechanical system, where displacement is the output and the applied force is input. $BC(s)$ with.

- A) 1×1 B) 1×2 **C) 2×1** D) 2×2

5) The inverse laplace $Y(s) = \frac{10(s+1)}{s^2+8s+15}$ is

- A) $Y(t) = 10e^{-5t} + 10e^{-4t}$ B) $Y(t) = 10e^{+5t} + 10e^{3t}$
C) $Y(t) = 10e^{5t} - 10e^{3t}$ **D) None**

6) Feed back is used As :-

- A) measure the output signal B) generate Feed back signal
C) determine the Feed back block Transfer Function **D) reduce the different between output and Input.**

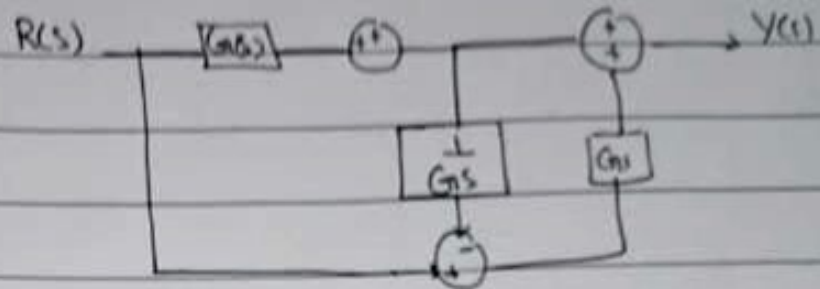


7) In speed control system the Feed back is

- A) pilot value B) control value
C) power cylinder **D) gear Box**

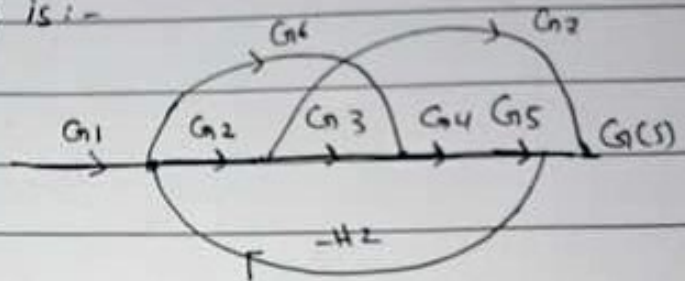
8) $X(s) / R(s)$

$G_1(s)$



9) number of non touching loop is:-

- A) 0 B) 4
C) 2 D) 2



10) $2\ddot{y}(t) + 4\dot{y}(t) = 24$

The Time constant is

- A) 2 B) 4 C) 0 **D) 0.5**

11) The steady state value is:

- A) 0 B) 1 C) 2 **D) 0.5**

12) For $\frac{C(s)}{R(s)} = \frac{1}{s^2 + 5s + 4}$, The unit step response is:-

- A) over damped B) under damped C) undamped
D) critical damped

13) The damping ratio is

- A) 1 **B) 0.25** C) 2 D) 4



Q2) $\ddot{x}(t) + 2\dot{x}(t) + 10x(t) = e^{-t}$

$\dot{x}(0) = 0, x(0) = 0$

Sol: $s^2 X(s) + 2sX(s) + 10X(s) = \frac{1}{s+1}$

$X(s)(s^2 + 2s + 10) = \frac{1}{s+1}$

$X(s) = \frac{1}{(s+1)(s^2 + 2s + 10)} = \frac{A}{s+1} + \frac{Bs+C}{s^2 + 2s + 10}$

$A(s^2 + 2s + 10) + Bs^2 + Bs + Cs + C = 1$

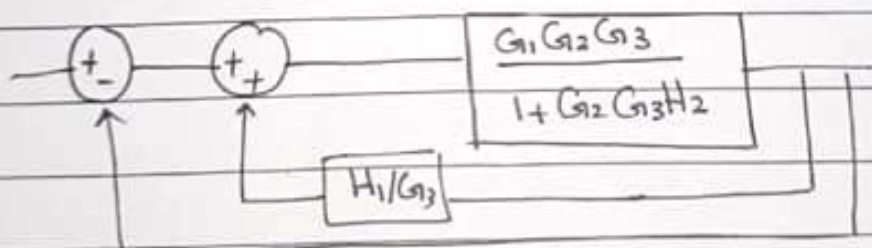
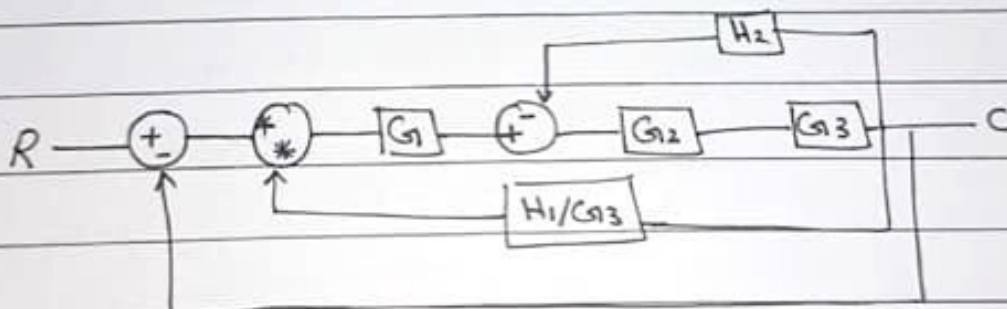
$As^2 + Bs^2 = 1 \rightarrow A + B = 0$ s^2 ko baad

$2A + Bs + Cs = 0 \rightarrow 0$ s ko baad

$10 + C = 1 \Rightarrow A = \frac{1}{9}, B = -\frac{1}{9}, C = -\frac{1}{9}$

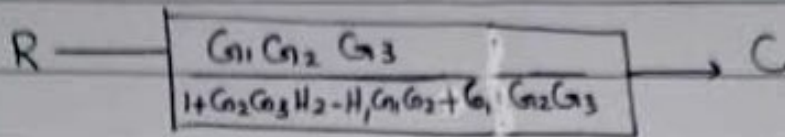
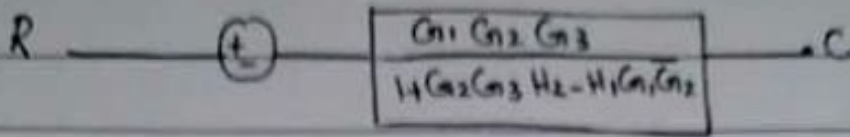
$= \frac{1}{9} + \frac{-\frac{1}{9}s - \frac{1}{9}}{s^2 + 2s + 10 + 1} = \frac{1}{9} e^{-t} + \frac{-\frac{1}{9}(s+1)}{(s+1)^2 + 9}$

$= \frac{1}{9} e^{-t} - \frac{1}{9} \cos 3t \cdot e^{-t}$ *





Power and Communication Engineering



Q1: select only the correct answer

1) the solution of the following transfer function

$$F(s) = \frac{s+2}{s(s+1)(s+3)}$$

26 marks

a) $f(t) = \frac{1}{6} - \frac{1}{2}e^{-t} + \frac{2}{3}e^{-3t}$

b. $f(t) = \frac{1}{2} - \frac{2}{3}e^{-t} + \frac{1}{6}e^{-3t}$

c. $f(t) = \frac{2}{3} - \frac{1}{2}e^{-t} + \frac{1}{6}e^{-3t}$

d. non of above

2) the following transfer function $F(s) = \frac{100}{(s^2+25)(s+2)}$ has

a) imaginary poles only

b. real poles only

c. imaginary and real poles

d. non of above

3) the following transfer function $\frac{2s+1}{s^2+4s+3}$ has

a) two poles and one zero

b. two zeros and one pole

c. one zero and repeated poles

d. non of ab

5) The differential equation from the following transfer function

$$\frac{X(s)}{F(s)} = \frac{7}{s^2+5s+10}$$

a)

$$7\frac{d^2x}{dt^2} + \frac{dx}{dt} + 10x = 5f.$$

b.

$$\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 10x = 7f.$$

c. $5\frac{d^2x}{dt^2} + \frac{dx}{dt} + 7x = 10f.$

d. non of ab

6) the transfer function from the following differential equation

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 3x = y(t)$$

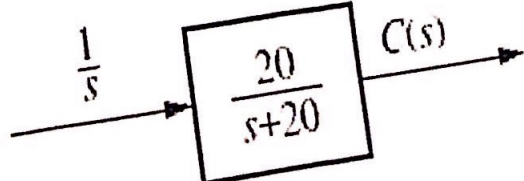
a) $\frac{X(s)}{Y(s)} = \frac{s^2+2s+3}{s}$

b. $\frac{Y(s)}{X(s)} = \frac{1}{s^2+2s+3}$

c. $\frac{X(s)}{Y(s)} = \frac{1}{s^2+2s+3}$

d. $\frac{X(s)}{Y(s)}$

Consider the following block diagram



7) the time constant is
a) 0.2

b. 0.11

c. 1/20

d. 20

8) the rise time is
a) 0.2

b. 0.11

c. 1/20

d. 20

8) the rise time is
a) 0.2

b. 0.11

c. 1/20

d. 20

10) $c(t) =$

a) $c(t) = e^{-5t} - 1$

b. $c(t) = e^{-5t}$

c. $c(t) = 1 - e^{-5t}$

d. none of the above

11) The following system

$$T(s) = \frac{5}{(s+3)(s+6)}$$

is

a underdamped

b. overdamped

c. critical damped

d. undamped

12) The following system

$$T(s) = \frac{(s+5)}{(s+10)^2}$$

is

a underdamped

b. overdamped

c. critical damped

d. undamped

12) The following system

$$T(s) = \frac{s+2}{s^2+9}$$

is

a underdamped

b. overdamped

c. critical damped

d. undamped

13) The following system

$$T(s) = \frac{20}{s^2 + 6s + 144}$$

is

a underdamped

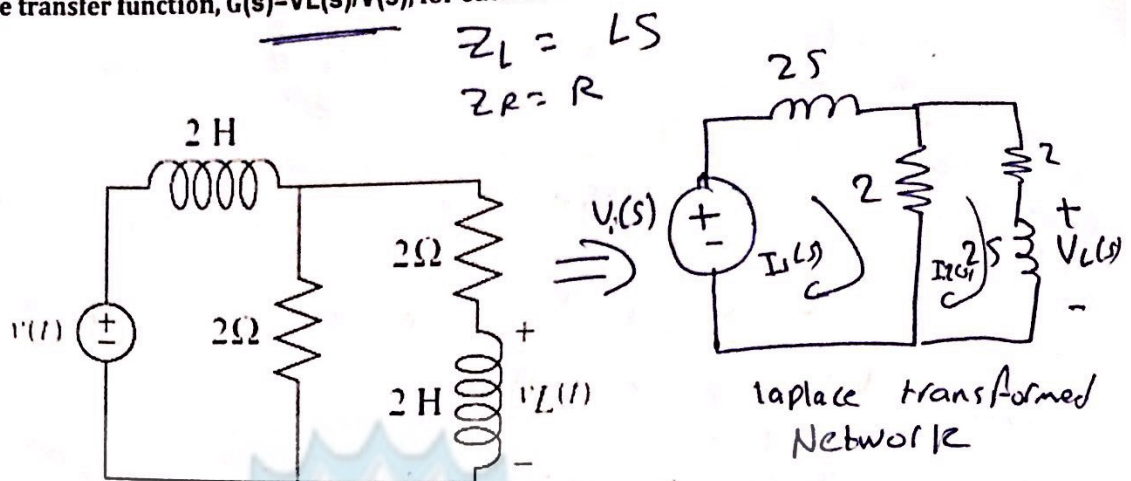
b. overdamped

c. critical damped

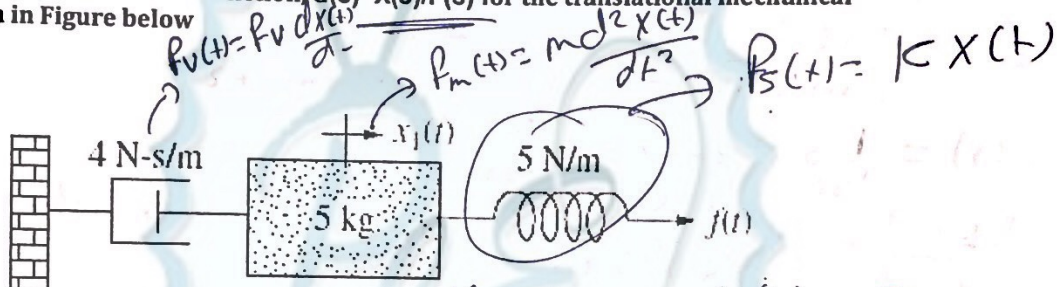
d. undamped

$$s_{1,2} = -3 \pm j35$$

Q2- 5 marks) Find the transfer function, $G(s) = V_L(s)/V(s)$, for each network shown in Figure below



Q3- 5 marks) Find the transfer function, $G(s) = X(s)/F(s)$ for the translational mechanical system shown in Figure below



$$F_o(t) = 5X(t) + 5 \frac{d^2 X(t)}{dt^2} + 4 \frac{dX(t)}{dt}$$

$$F_o(s) = 5s^2 X(s) + 4s X(s) + 5 X(s) = (5s^2 + 4s + 5) X(s)$$

$$G(s) = \frac{X(s)}{F_o(s)} = \frac{1}{5s^2 + 4s + 5}$$

5-4 marks) Given the transfer function

$$G(s) = \frac{b}{s^2 + as + b}$$

Evaluate percent overshoot, settling time, peak time, and rise time for the following values: $a = 4$; $b = 25$. Also, plot the poles.

$$W_n = \sqrt{b} = \sqrt{25} = 5$$

$$s = \frac{-a \pm \sqrt{a^2 - 4b}}{2} = \frac{-4 \pm \sqrt{16 - 100}}{2} = -2 \pm j\sqrt{21}$$

$$\zeta = \frac{|a|}{2W_n} = \frac{4}{2 \times 5} = \frac{4}{10} = 0.4$$

$$\text{Percent overshoot} = \% OS = \frac{C_{max} - C_{final}}{C_{final}} \times 100$$

$$C(t) = C_{max} = 1 + e^{-(\zeta \pi / \sqrt{1 - \zeta^2})}, \quad C_{final} = 1$$

$$= 1.2538$$

$$\% OS = \frac{1.2538 - 1}{1} \times 100 = 25.3826\%$$

$$\text{Settling time} = \frac{4}{\zeta W_n} = \frac{4}{0.4 \times 5} = 2 \text{ seconds}$$

$$\text{Peak time} = T_p = \frac{\pi}{W_n \sqrt{1 - \zeta^2}} = \frac{\pi}{5 \sqrt{1 - 0.16}} = 0.68 \text{ second}$$



أسئلة سنوات سابقة

المادة: نظرية التحكم

Q1:- Multiple choice

$$1) F(s) = \frac{s+2}{s(s+1)(s+3)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+3}$$

$$s+2 = A(s+1)(s+3) + B(s+3)(s) + C(s+1)(s)$$

$$s=0 \rightarrow A = 2/3$$

$$s=-1 \rightarrow B = -1/3, \quad s=-3 \rightarrow C = -1/6$$

$$f(t) = 2/3 - 1/3 e^{-t} - 1/6 e^{-3t}$$

(C)

$$(2) F(s) = \frac{100}{(s^2+25)(s+2)} \text{ has: -}$$

$$s_{1,2} = \pm j5, \quad s_3 = -2$$

Has Imaginary and real poles

(C)

$$(3) F(s) = \frac{2s+1}{s^2+4s+3}$$

$$\text{Zeros: - } 2s+1=0 \rightarrow s = -1/2$$

$$\text{Poles: - } s^2+4s+3=0 \rightarrow s_1=-1, s_2=-3$$

It has two poles and one zero

(A)

$$(u) \frac{X(s)}{F(s)} = \frac{7}{s^2+5s+10}$$

take laplace inverse for both sides

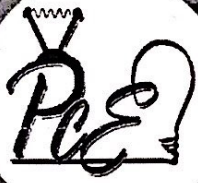
$$s^2 \rightarrow \frac{d^2}{dt^2}, \quad s \rightarrow \frac{d}{dt}$$

$$\frac{d^2 x}{dt^2} + 5 \frac{dx}{dt} + 10x = 7f$$

$$X(s) \rightarrow x(t) \\ F(s) \rightarrow f(t)$$

(B)

Don't miss the change ... Be part of it



أسئلة سنوات سابقة

المادة: نظرية التحكم

(5) $\frac{d^2 x}{dt^2} + 2 \frac{dx}{dt} + 3x = y(t)$
 $\swarrow$ output $\searrow$ input

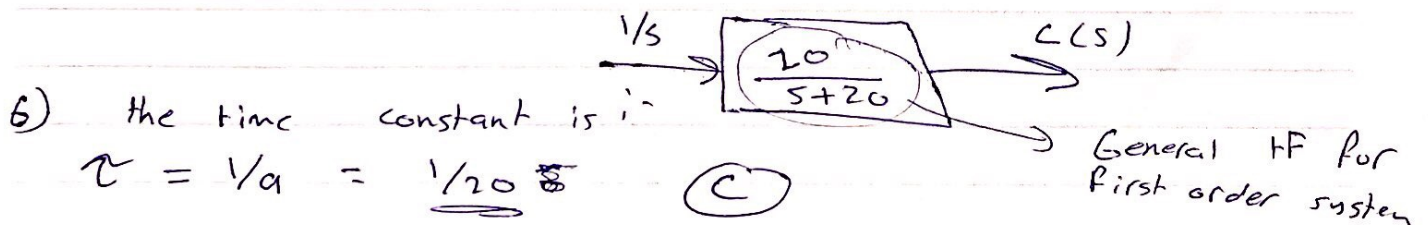
take laplace for both sides: $\frac{d^2}{dt^2} \rightarrow s^2$

$s^2 X(s) + 2sX(s) + 3X(s) = Y(s)$ $\frac{d}{dt} \rightarrow s$, $X \rightarrow X(s)$
 output $\leftarrow X(s)(s^2 + 2s + 3) = Y(s) \rightarrow$ input $y(t) = Y(s)$

transfer function = $\frac{\text{output}}{\text{input}} = \frac{X(s)}{Y(s)} = \frac{1}{s^2 + 2s + 3}$

(C)

Consider the block diagram for question: 6, 7, 8, 9



(7) The rise time is:-

$t_r = \frac{2.2}{a} = \frac{2.2}{20} = 0.11$ (B)

(8) $C(t) = ?$

For the general T.F for first order system

$C(t) = 1 - e^{-at} = 1 - e^{-20t}$

(D)



أسئلة سنوات سابقة

المادة:

11) The following system $T(s) = \frac{5}{(s+3)(s+6)}$ is:

Sol:- The poles of the system are $(s+3)(s+6)=0$
 $s_1 = -3, s_2 = -6$ as there are two real and different poles the system is overdamped. (B)

12) The following system $T(s) = \frac{(s+5)}{(s+10)^2}$ is:

Sol:-

The poles of the system are $(s+10)^2=0$
 $s_{1,2} = -10$ so the system is critically damped (C)

13) The following system $T(s) = \frac{s+2}{s^2+9}$ is:

Sol:-

The poles of the system are $s^2+9=0 \Rightarrow s_{1,2} = \pm j3$
The system is undamped system. (D)

(14) The following system

$$T(s) = \frac{20}{s^2+6s+144} \text{ is:-}$$

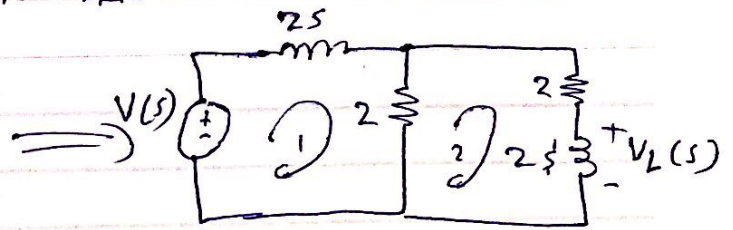
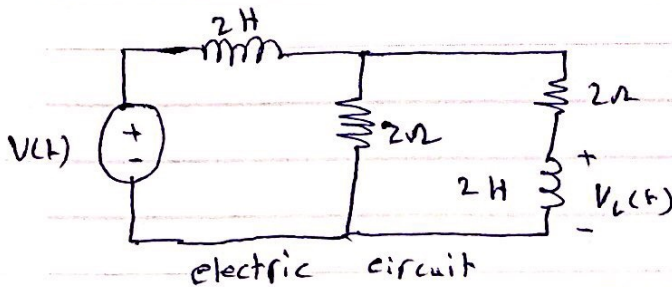
The poles of this system is $s^2+6s+144=0$
 $s_{1,2} = -3 \pm j3\sqrt{15}$ so the system is underdamped (A)



أسئلة سنوات سابقة

المادة:

Q2) Find the transfer function $G(s) = V_L(s)/V(s)$ for the network shown in figure below



$$Z_L(s) = Ls$$

$$Z_R(s) = R$$

mesh ①:-

$$I_1(s) (2 + 2s) - I_2(s) (2) = V(s) \quad \text{--- (1)}$$

mesh ②:-

$$-I_1(s)(2) + I_2(s) (4 + 2s) = 0 \quad \text{--- (2)}$$

$$\textcircled{2} \times (s+1) + \textcircled{1}$$

$$\Rightarrow$$

$$-I_2(s)(2) + I_2(s) (4 + 2s')(s'+1) = V(s)$$

$$-2I_2(s) + I_2(s) (2s'^2 + 6s' + 4) = V(s)$$

$$I_2(s) (2s'^2 + 6s' + 2) = V(s)$$

$$V_L(s) = 2s' \times I_2(s) \Rightarrow I_2(s) = \frac{V_L(s)}{2s'}$$

$$\frac{V_L(s)}{2s'} (2s'^2 + 6s' + 2) = V(s)$$

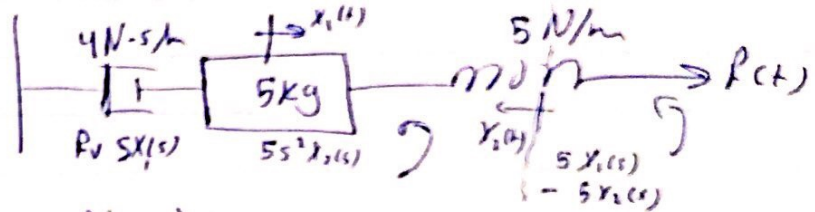
$$\frac{V_L(s)}{V(s)} = \frac{s'}{s'^2 + 3s' + 1}$$



أسئلة سنوات سابقة

المادة: نظرية التحكم

Q3:-



Sol:-

The equations of motion:-

$$(5s^2 + 4s + 5) X_1(s) - 5 X_2(s) = 0 \quad (1)$$

$$-5 X_1(s) + 5 X_2(s) = F(s) \quad (2)$$

$$(1) + (2) \Rightarrow (5s^2 + 4s) X_1(s) = F(s)$$

$$\text{Transfer function} = \frac{X_1(s)}{F(s)} = \frac{1}{5s^2 + 4s}$$

Q4:-

Sol:-

$$\omega_n = \sqrt{b} = \sqrt{25} = 5$$

$$\zeta = \frac{1a}{2\omega_n} = \frac{4}{2 \times 5} = 0.4$$

$$\text{Percent over shoot} = \% OS = \frac{c_{\max} - c_{\text{final}}}{c_{\text{final}}} \times 100; c_{\text{final}} = 1$$

$$c_{\max} = 1 + e^{-(\zeta \pi / \sqrt{1-\zeta^2})}$$

$$= 1.2538$$

$$\% OS = \frac{1.2538 - 1}{1} \times 100\% = 25.3826\%$$

$$\text{Settling time} = \frac{4}{\zeta \omega_n} = \frac{4}{0.4 \times 5} = 2 \text{ seconds}$$

$$\text{Peak time} = T_P = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \frac{\pi}{5 \sqrt{1-0.16}} = 0.68 \text{ seconds}$$

Rise time = time needed to rise

from 0.1 to 0.9

$$c(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta \omega_n t} \cos(\omega_n \sqrt{1-\zeta^2} t - \tan^{-1}(\zeta / \sqrt{1-\zeta^2}))$$

$$c(t) = 1 - 1.091 e^{-2t} \cos(4.5825t - 23.57^\circ)$$

$$c(t_1) = 0.1; t_1 = 0.09625$$

$$c(t_2) = 0.9; t_2 = 1.19289 \quad t_r = t_2 - t_1 = 1.0975$$

Don't miss the change ... Be part of it

A control system is given as

$$\dot{x}_1'(t) = x_1(t) + x_2(t) + u_1(t) + u_2(t)$$

$$\dot{x}_2'(t) = x_1(t) + u_1(t)$$

$$y_1(t) = x_1(t) + u_2(t)$$

$$y_2(t) = u_1(t) + u_2(t)$$

Answer 12 and 13

12. the state matrix A is

$$A \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$B \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$C \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$D \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

13. the output matrix D is

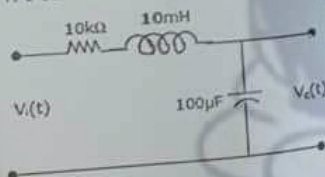
$$A \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$B \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$C \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

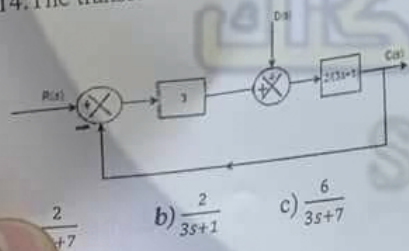
$$D \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

14. For the circuit shown in the Figure below, the initial conditions are zero. Its transfer function $H(s) = V_c/V_i$



- A) $\frac{1}{s^2 + 10^6 s + 10^6}$
 B) $\frac{10^6}{s^2 + 10^3 s + 10^6}$
 C) $\frac{10^3}{s^2 + 10^3 s + 10^6}$
 D) $\frac{10^6}{s^2 + 10^6 s + 10^6}$
 E) None of these

14. The transfer function from D(s) to Y(s) is :



$$\frac{2}{s+7}$$

$$b) \frac{2}{3s+1}$$

$$c) \frac{6}{3s+7}$$

$$d) \frac{2}{3s+6}$$

Select the most correct answer from the alternatives given, and then write it in the table below.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	B	C	D	B	B	C	D	D	B	X	B	D	D	C

1. For closed control system which of the following statements is correct?

- (A) Less expensive
(B) May be unstable stable
(C) Construction is simple and maintenance easy
(D) Errors are caused by disturbances

2. The first control system developed is

- (A) Temperature control system (B) level control system
(C) speed control system (D) robot system

3. Transfer function of a system is used to calculate which of the following?

- (A) The order of the system (B) The time constant
(C) The output for any given input (D) The steady state gain

4. In temperature control system, the feedback path is

- (A) Thermometer (B) Analog to digital converter (C) Interface (D) All above

5. What should be the value of Laplace transform for the time-domain signal function?

$f(t) = e^{-at} \sin(\omega t) \cdot u(t)$

- (A) $1/(s+a)$ (B) $\omega / \{(s+a)^2 + \omega^2\}$ (C) $(s+a) / \{(s+a)^2 + \omega^2\}$ (D) $A\omega / (s^2 + \omega^2)$

6. When the function $F(s)$ has multiple poles, the type of the function $f(t)$ is

- (A) Exponential (B) damped ramp (C) undamped (D) critically damped sinusoidal

7. The inverse Laplace transform of $F(s) = 2/(s+1)(s+2)$ is

- (A) $2e^{-t} - 2e^{-2t}$ (B) $2e^{-t} + 2e^{-2t}$ (C) $2e^{-t} - 2e^{-2t}$ (D) $2e^{-t} + 2e^{-2t}$

8. The system given by $y''(t) + ay'(t) + y(t) = x(t)$, where a constant, can be represented by

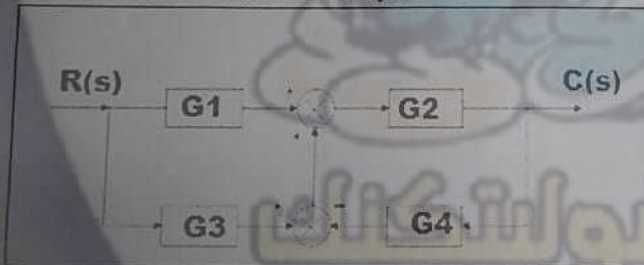
- (A) simulation model (B) transfer function (C) state space model (D) all mentioned

9. The transfer function of RL circuit, where the applied voltage is the input and the current is the output is

- (A) $\frac{Ls}{RLs+1}$ (B) $\frac{s}{RLs+1}$ (C) $\frac{L}{Ls+1}$ (D) $\frac{1}{Ls+R}$

10. Consider the block diagram shown,

If the transfer function of the system is

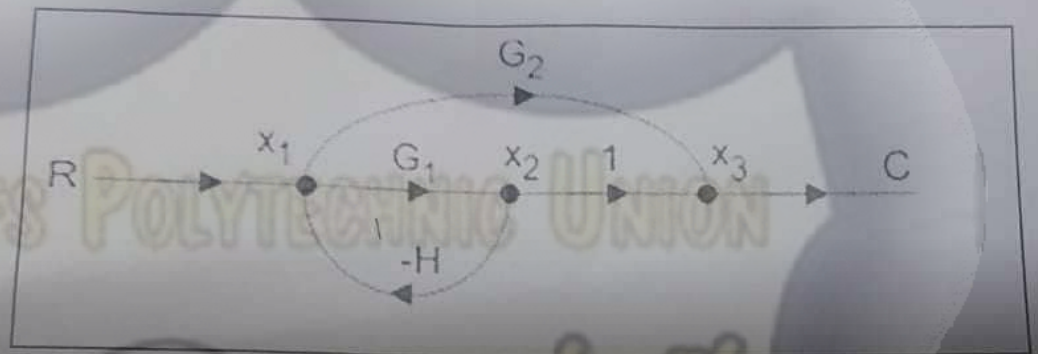


given by $T(s) = G1G2 + G2G3/1 + X$. Then X is:

- (A) $G2G3G4$ (B) $G2G4$ (C) $G1G2G4$ (D) $G3G4$

A node which has both incoming and outgoing branches is called a/an
 (A) Input node (B) mixed node (C) output node (D) general node

13. Use Mason formula to calculate the transfer function of the following figure



- (A) $G1/(1+G2H)$ (B) $G2/(1+G1H)$
 (C) $(G1+G2)/(1+G1H)$ (D) None

14. In the given state space equations

$$\dot{x}_1(t) = x_1(t) + x_2(t) + u_1(t) + u_2(t) \quad \dot{x}_2(t) = x_1(t) + u_1(t) \quad \dot{x}_3(t) = x_2(t)$$

the state matrix A is

- (A) $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ (B) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (C) $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$ (D) None

15. for a standard translation mechanical system, The input signal to the system
 (A) displacement (B) velocity (C) applied force (D) acceleration

P
 • CO

13)

$P_1 < G_1$

SPU